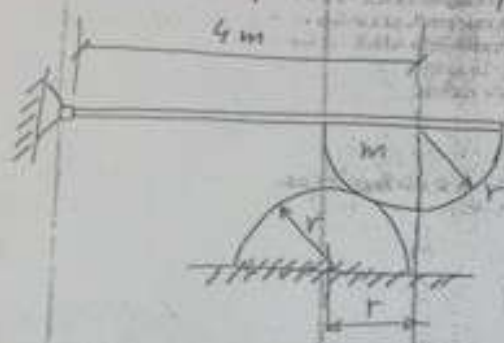


Izpit iz Mehanike I

NTF, IM, 99 2016

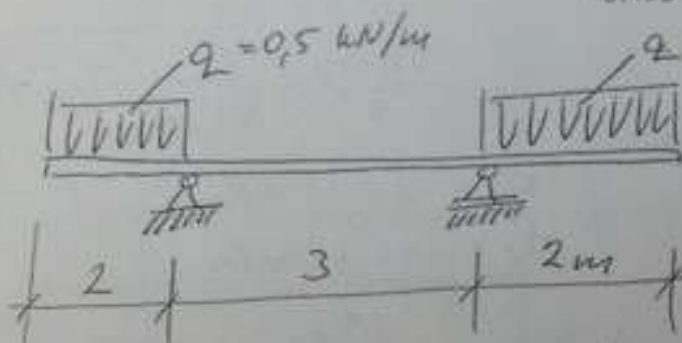
- 4) Izračunajte velikosti reakcijskih sil!
Masa droga je zanemarljiva, trenja ni.



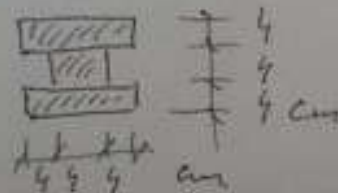
$$r = 1,2 \text{ m}$$

$$m = 280 \text{ kg}$$

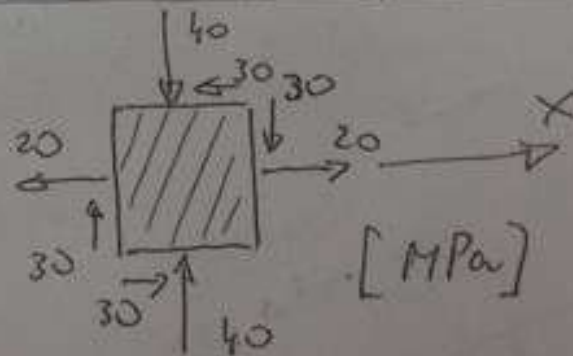
- 2) Narišite diagrame N, T, M ! Izračunajte M_{max} !
Izračunajte elastične napet, posledica upogiba



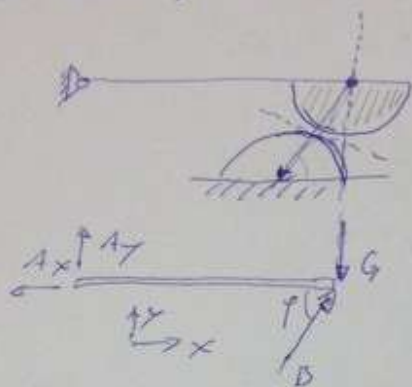
prerez



- 3) V delčku stene je znano napetostno stanje:
Določite velikosti glavnih napetosti in smeri, pri katerih se pojavijo!



1) $G = m \cdot g = 280 \cdot 9,81 = 2747 \text{ N}$

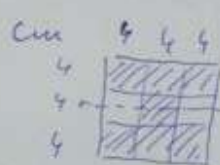
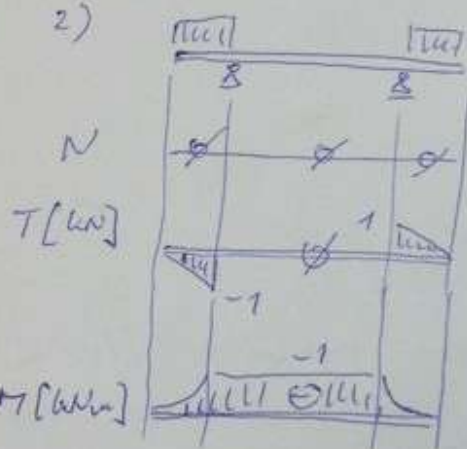


$\rho = \arccos \frac{r}{2r} = 60^\circ$

A: $-G \cdot 4 + B \cdot \sin 60^\circ \cdot 4 = 0$

$$\begin{cases} B = 3172 \text{ N} \\ A_x = 1586 \text{ N} \\ A_y = 0 \end{cases}$$

2)



$J_z = \frac{12 \cdot 12^3}{12} - \frac{8 \cdot 4^3}{12}$

$J_z = 1685,3 \text{ cm}^4$

$\sigma_x = \frac{100 \cdot (\pm 6)}{1685,3} = \pm 0,356 \text{ kN/cm}^2$

3)

$\sigma_{ij} = \begin{pmatrix} 20 & -30 & 0 \\ -30 & -40 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ MPa}$

$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$\sigma_1 = 32,42 \text{ MPa}$

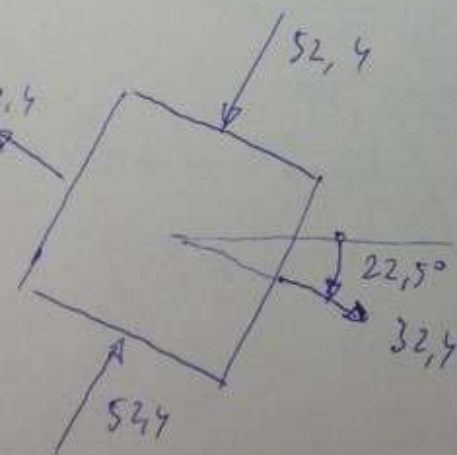
$\sigma_2 = -52,42 \text{ MPa}$

$\tan(2\alpha) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} ; \alpha = -22,5^\circ$

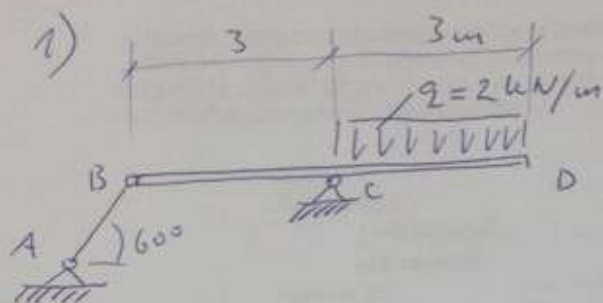
Kontrol:

$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos(2\alpha) + \tau_{xy} \sin(2\alpha)$

$\sigma_n(\alpha = -22,5^\circ) = 32,42 \text{ MPa}$



izpit iz Mehanike I
NTF, UN - OGRO
19.5.2016

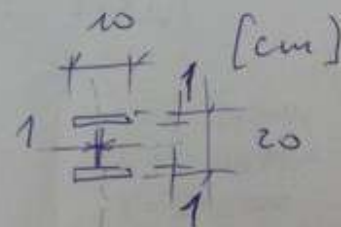
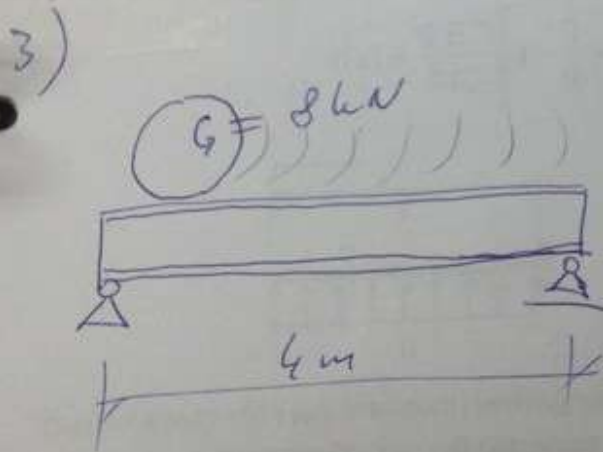


$N, T, M = ?$
 M_{elast}

2)

$$\sigma_{ij} = \begin{pmatrix} 50 & 30 & 0 \\ -10 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ MPa}$$

Mohrova krožnica?
Glavne napetosti?
Koti med normalami
ploskev glavnih napetosti in
 x, y, z koordinatni sistem?

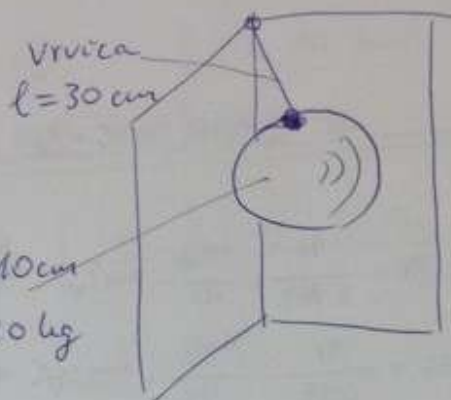


Elastične nap., ki
so posledica upogiba?
Velikost, mesto na nosilcu

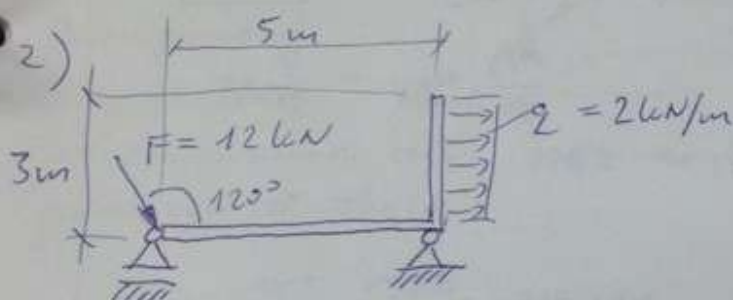
Izpit iz Mehanike I

NTF - UN - OGRD 1.6.2015

- 1) Izračunajte veličnosti reakcij in silo v vrvici! Trenja ni.

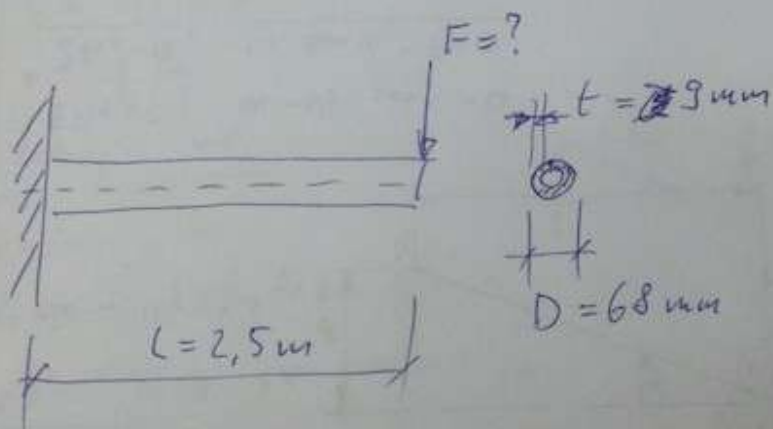


krogla, $r = 10 \text{ cm}$
 $m = 10 \text{ kg}$



$N, T, M = ?$

3)

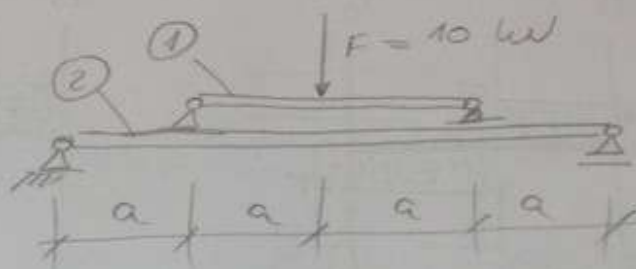


$$\sigma_{\text{dop}} = 160 \text{ N/mm}^2$$

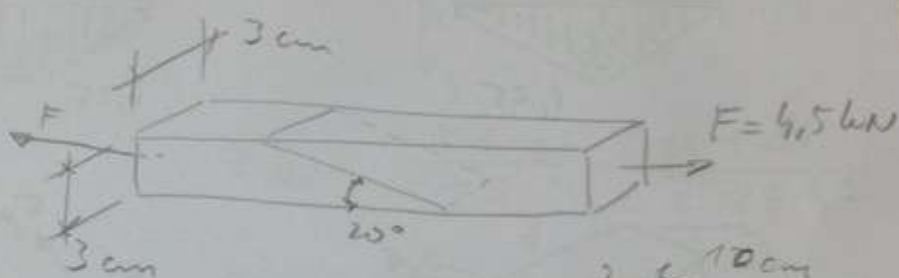
Koliko se
sme biti sila
 F , da napetost
v nosilcu, ki
je posledica
upogiba, doseže
 $\sigma_{\text{dop}}?$

Krajcar
 Gorja Dobovska Izpit iz mehanike
 UTF, 4.5.2016

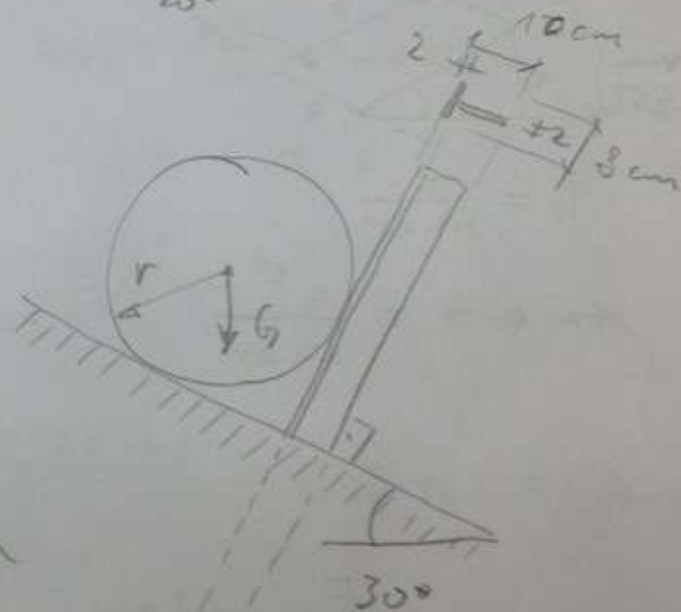
- 1) $N, T, M = ?$
 v ① in ②
 $a = 1,25\text{ m}$



- 2) Izračunajte
 komponenti
 napetostnega
 vektora v
 prerezu A-A

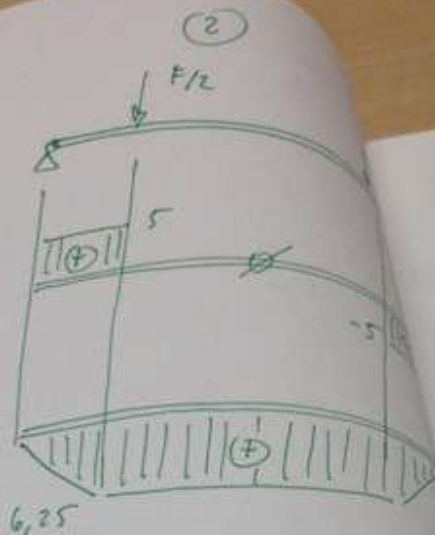
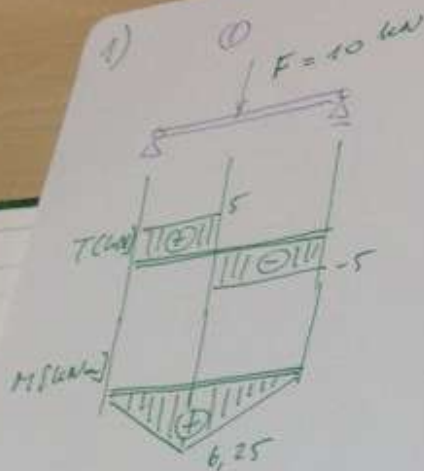


- 3) $G = 12\text{ kN}$
 $r = 0,3\text{ m}$
 trenja ni



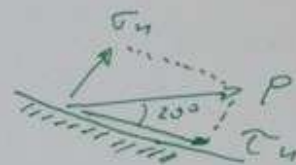
Izračunajte
 velikost elastomernih
 napetosti, ki so posledica
 upogiba. Prikažite to (ta)

места



$$b = \frac{a}{\sin 20^\circ}$$

$$A = a \cdot b = \frac{a^2}{\sin 20^\circ}$$



$$p = \frac{F}{A}$$

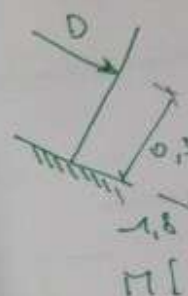
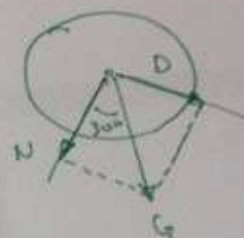
$$\sigma_n = p \cdot \sin 20^\circ$$

$$\tau_n = p \cdot \cos 20^\circ$$

$$\sigma_n = \frac{F}{a^2} \sin^2 20^\circ$$

$$= \frac{4.5}{9} \cdot 0.117 = 0.05848 \frac{\text{kN}}{\text{cm}^2}$$

$$\tau_n = \frac{F}{a^2} \sin 20^\circ \cdot \cos 20^\circ = 0.1607 \frac{\text{kN}}{\text{cm}^2}$$

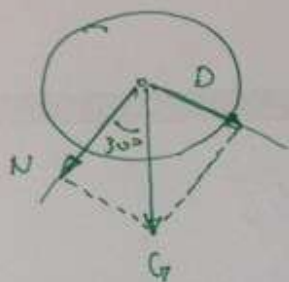


$$m = -$$

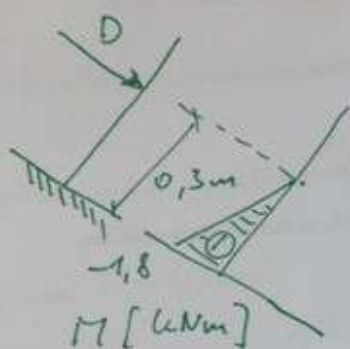
$$c =$$

Warning: IllegalMediasize
PCL XL error

normale fälle B.

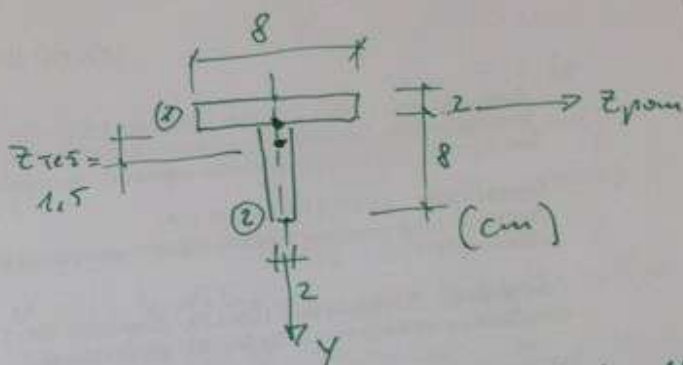


$$D = G \cdot \sin 30^\circ = 12 \cdot 0,5 = 6 \text{ kN}$$



$$M = -1,8 \text{ kNm}$$

$$\sigma = \frac{M}{J} y$$



$$\bar{z}_{\text{res}} = \frac{\sum A_i \bar{z}_i}{\sum A_i} = \frac{16 \cdot 1 - 16 \cdot 4}{2 \cdot 16}$$

$$\bar{z}_{\text{res}} = -\frac{3}{2} = -1,5 \text{ cm}$$

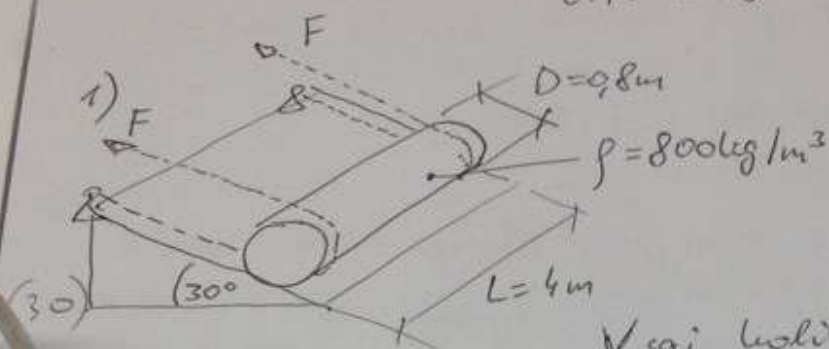
$$J_z = \frac{2 \cdot 8^3}{12} + 2,5^2 \cdot 16 + \frac{8 \cdot 2^3}{12} + 2,5^2 \cdot 16$$

$$J_z = 290,6 \text{ cm}^4$$

$$\sigma_{zs} = \frac{-180 \text{ kNm} \cdot (-3,5 \text{ cm})}{290,6 \text{ cm}^4} = +2,167 \text{ kN/cm}^2$$

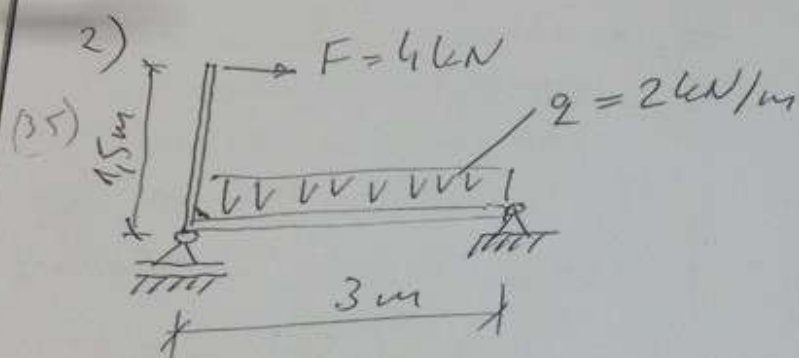
$$\sigma_{sp} = \frac{-180 (+6,5)}{290,6} = -4,025 \text{ kN/cm}^2$$

Izpit iz mehanike
inženirstvo materialov
6.7. 2016

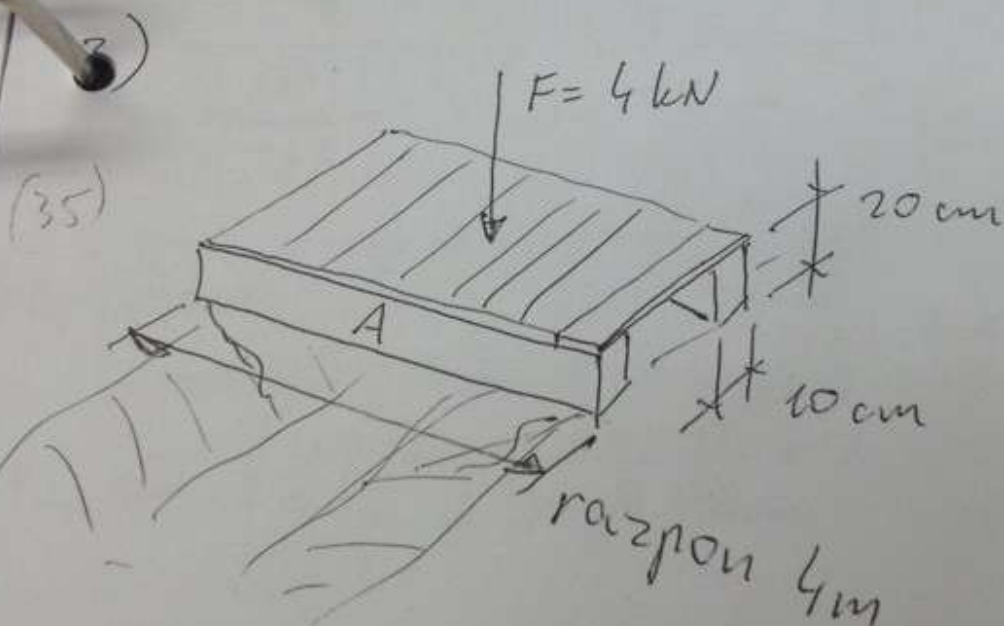


Kupujete vrv,
s katero želite
premakniti valj
po klancu navzgor
brez sunkov.

Vsaj kolikšno pretržno silo
mora vrv zdržati, če je koef. varnosti 6?



Diagrami
 N, T, M ?

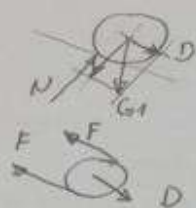


Kolikšna je
elastična napetost
v nosilcu A, ki je
posledica upogib
nosilca zaradi
obremenitve F

4) Teža celotila: $V = \frac{0,8 \cdot \pi}{4} \cdot 4 = 2,51 \text{ m}^3$

$G = \rho \cdot V \cdot g = 800 \cdot 2,51 \cdot 10 = 16080 \text{ N}$

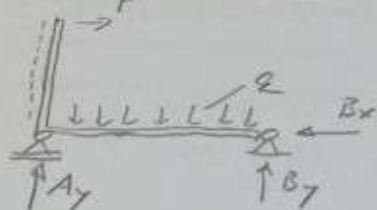
na eno vrh pade pol obtežbe $G_e = G/2 = 8042 \text{ N}$



$D = G_e \cdot \sin 30^\circ = 4021 \text{ N}$

$F = D/2 = 2011 \text{ N}$, Pretrežna sila vsaj
 $F_{\text{pretr}} = 6 \cdot F = 12,1 \text{ kN}$

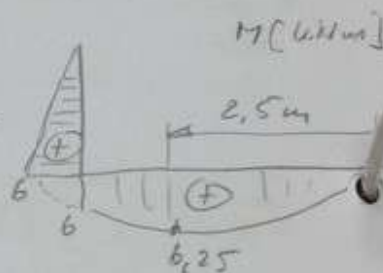
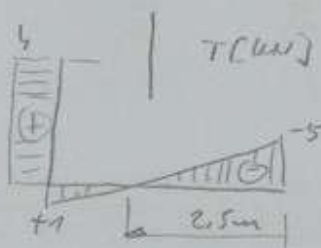
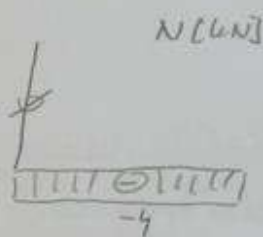
1)



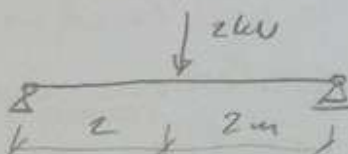
$B_x = 4 \text{ kN}$

$A_y = 1 \text{ kN}$

$B_y = 5 \text{ kN}$



3)



$J = \frac{10 \cdot 20^3}{12} = 6666,6 \text{ cm}^4$

$W = \frac{J}{10} = 666,6 \text{ cm}^3$

$\sigma = \frac{200 \text{ kN cm}}{666,6 \text{ cm}^3} = 0,3 \text{ kN/cm}^2$

1) Lestev, ta
4 m, je
30 steno (t
mora biti
dostvijo
neke 10
3 m p

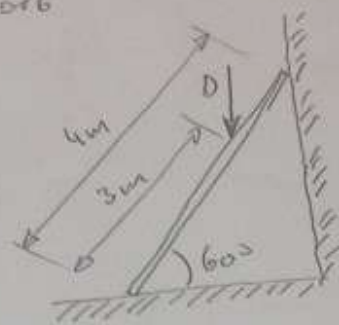
2) Izračun
elektro
posle
35 naravn

3) k
v

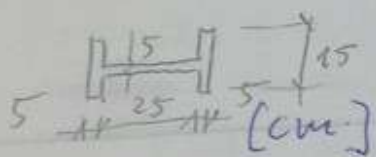
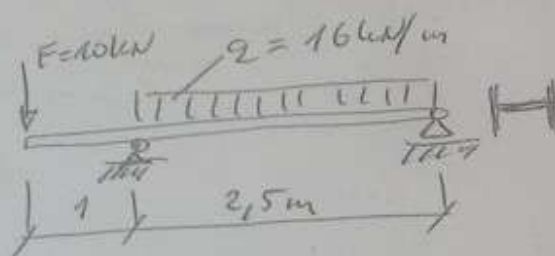
35

Izpit iz mehanike I
NTF, IM, 27.6.2016

- 1) Lestev, teže 150 N in dolžine 4 m , je naslonjena na gladko steno (trenja ni). Vsaj kolikšen mora biti koef. trenja med lestvijo in podlago, da delavec, teže 1000 N , lahko povzpe 3 m po lestvi?



- 2) Izračunajte velikosti in ekstremnih napetosti, ki so posledica upogiba! Narišite diagram M !

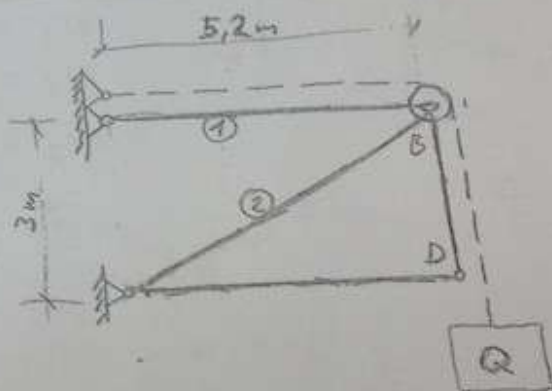


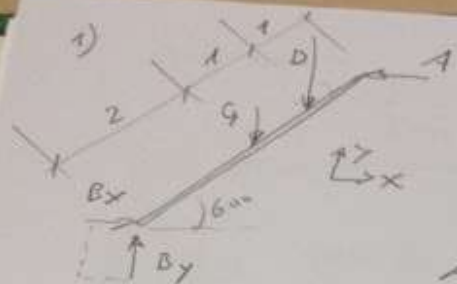
- 3) Izračunajte minimalne velikosti preozov palic 1 in 2, tako, da ne bo presežena dopustna napetost $\sigma_{dop} = \pm 70\text{ MPa}$!

Izračunajte gravite točke D (x, y)!

$E = 0,7 \cdot 10^5\text{ MPa}$, $Q = 60\text{ kN}$

Trenja ni! Ne upoštevajte uklona! Dimenzije kolena v točki B zanemarite!



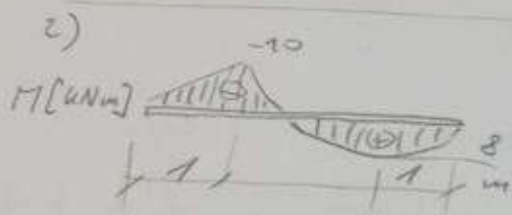


$$\begin{aligned}\sum F_{ix} &= 0; & A &= B_x \\ \sum F_{iy} &= 0; & B_y &= q + D = 1150 \text{ N} \\ \sum M_{iB} &= 0; & A \cdot l \cdot \sin 60^\circ - q \cdot \frac{l}{2} \cos 60^\circ - D \cdot \frac{3}{4} l \cdot \cos 60^\circ &= 0\end{aligned}$$

$$A = 426,13 \text{ N} = B_x$$

$$B_y = 11, \quad B_x = T, \quad T = f \cdot N$$

$$f = \frac{B_x}{B_y} = 0,4142$$



$$M_{elast} = 10 \text{ kNm}$$

$$I = 2 \cdot \frac{5 \cdot 15^3}{12} + \frac{25 \cdot 5^3}{12} = 3072,916 \text{ cm}^4$$

$$\sigma = \frac{M \cdot e}{J} = \frac{1000 \cdot 7,5}{3072,916} = 2,44 \text{ kN/cm}^2$$



$$-N_1 - N_2 \cdot \cos 30^\circ - S = 0$$

$$-N_2 \cdot \sin 30^\circ - Q = 0$$

$$N_2 = -120 \text{ kN (tlač)}$$

$$N_1 = 43,92 \text{ kN (ťah)}$$

$$N_3 = N_4 = 0$$

$$A_1 = \frac{43,92}{7} = 6,275 \text{ cm}^2$$

$$A_2 = \frac{-120}{-7} = 17,14 \text{ cm}^2$$

$$\Delta l_1 = l_1 \cdot \frac{\sigma_1}{E} = 5200 \cdot \frac{20}{97 \cdot 10^5} = 5,2 \text{ mm}$$

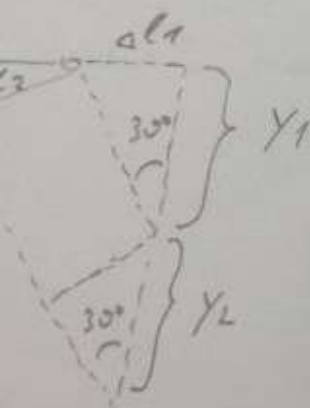
$$\Delta l_2 = 6003,33 \cdot \frac{-70}{20000} = -6,003 \text{ mm}$$

$$\tan 30^\circ = \frac{\Delta l_1}{y_1}; \quad y_1 = 9 \text{ mm}$$

$$\sin 30^\circ = \frac{|\Delta l_2|}{y_2}; \quad y_2 = 12 \text{ mm}$$

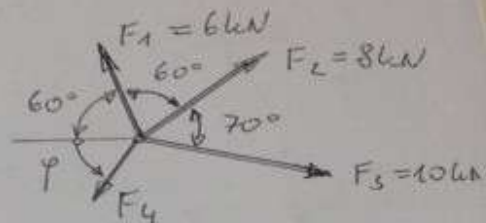
$$y_D = 9 + 12 = 21 \text{ mm}$$

$$x_D = 0$$

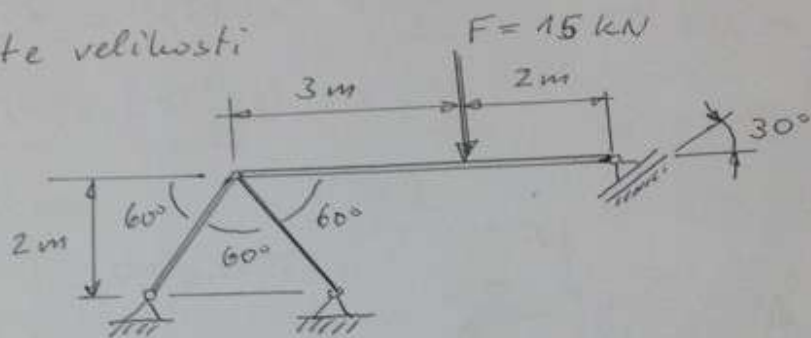


1. Kolokvij iz Osnov mehanike
24.11.2016

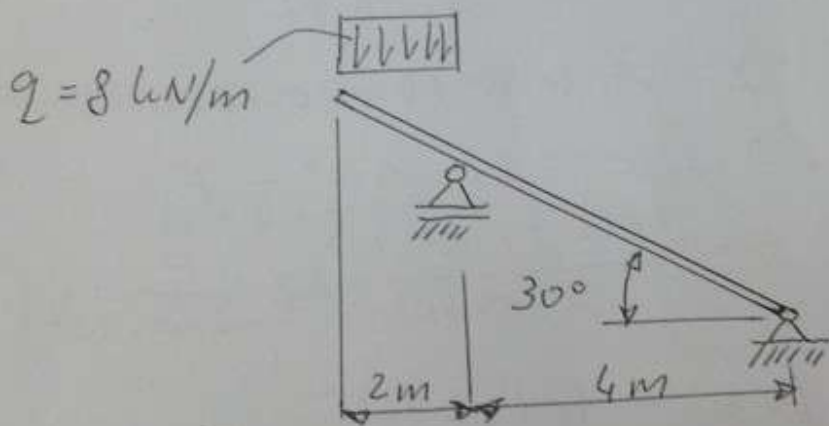
- 1) S kolikošno pakostjo in pod kakšnim kotom φ mora delovati sila F_4 , da bo doseženo ravnotežje?



- 2) Izračunajte velikosti reakcij!



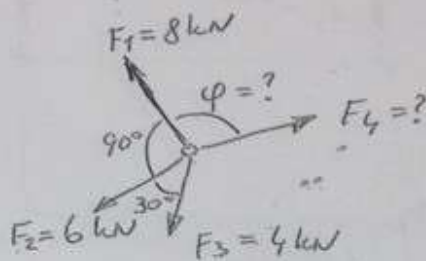
- 3) Izračunajte N, T, M



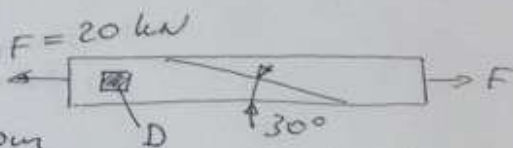
Osnove mehanike

NTF IM UN 24.8.2016

- 1) S kolikošno jakostjo
(30) ter pod kakšnim kotom φ
mora delovati sila F_4 ,
da bo doseženo ravnotežje?



- 2) Palica prereza 2×2 cm
je zlepljena tako, da je
ravnina lepljenja pod kotom



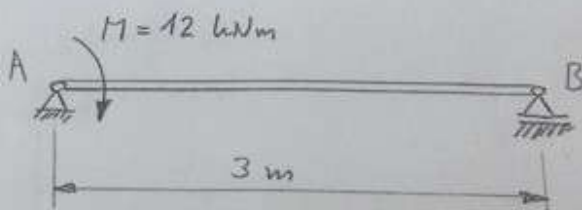
- (35) 30° glede na vzdolžno os palice. Določite komp.
nap. tenz. za delec D (pazite orientacijo delca)
ter izračunajte komponenti napetostnega
vektora na ravnini lepljenja

- 3) Določite:

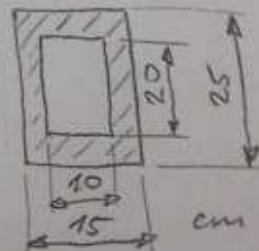
- a) mesto in velikost
(35) največjih upogibnih
napetosti,

- b) mesto in velikost
največjega povzega!

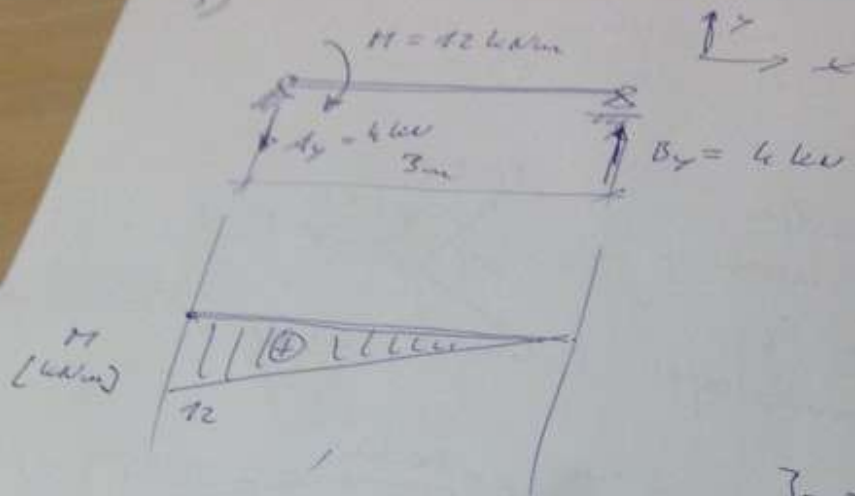
$E = 1000 \text{ kN/cm}^2$



prerez nosilca



3)



$$\sigma_x = \frac{1200 \text{ kN cm}}{1029,2 \text{ cm}^3}$$

$$\sigma_x = \pm 1,166 \text{ kN/cm}^2$$

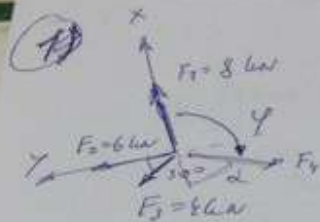
$$J_z = \frac{15 \cdot 25^3}{12}$$

$$J_z = 12865 \text{ cm}^4$$

$$W_z = \frac{J_z}{12,5} = 1029,2$$

Kolikšna
pod ka
ra delova
bo dose
Palica
je zlepr
ravnina
30° gled
vap. ter
ter izr
vektor

- Določiti
- mesto
 - največ
napetost
 - mesto
najtež
E =



$$R_x = F_1 - F_3 \cdot \sin 30^\circ = 6 \text{ kN}$$

$$R_y = F_2 + F_3 \cdot \cos 30^\circ = 9,464 \text{ kN}$$

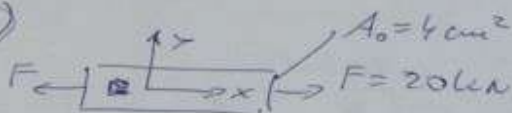
$$\vec{F}_4 = (-R_x, -R_y) = (-6, -9,464) \text{ kN}$$

$$F_4 = \sqrt{F_{4x}^2 + F_{4y}^2} = 11,206 \text{ kN}$$

$$\tan \alpha = \left| \frac{F_{4y}}{F_{4x}} \right| \Rightarrow \alpha = 57,626^\circ$$

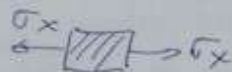
$$\varphi = 180^\circ - \alpha = 122,374^\circ$$

2)



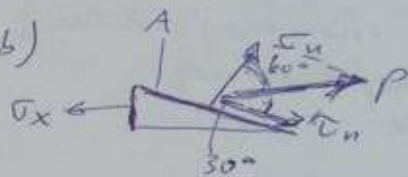
$$\sigma_x = \frac{20}{4} = 5 \text{ kN/cm}^2$$

2a) delec D:



$$\sigma_{ij} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{ kN/cm}^2$$

2b)



$$\text{mag. velut } p: p = \frac{F}{A}$$

$$p = 2,5 \text{ kN/cm}^2$$

Alt:

$$\frac{2}{\sin 30^\circ} = 4$$

$$\sin 30^\circ = \frac{2}{l}$$

$$l = \frac{2}{\sin 30^\circ} = 4 \text{ cm}$$

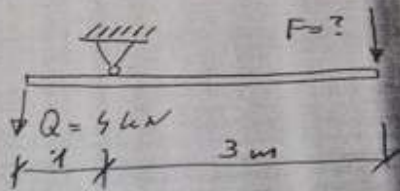
$$A = 2 \cdot 4 = 8 \text{ cm}^2$$

$$\sigma_n = p \cdot \cos 60^\circ = 1,25 \text{ kN/cm}^2$$

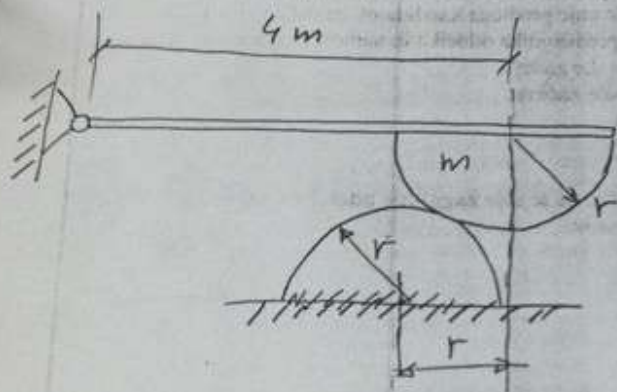
$$\tau_n = p \cdot \sin 60^\circ = 2,165 \text{ kN/cm}^2$$

Izvit iz Mehanike - Statika
 Mehanika - statika 06

- 1) Izračunajte jakost
 sile F , da bo sistem
 v ravnovesju!
 Narišite diagrame
 N, T, M !



- 2) Izračunajte velikosti reakcijskih sil!
 Masa droga je zanemarljiva, trenja ni.



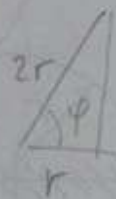
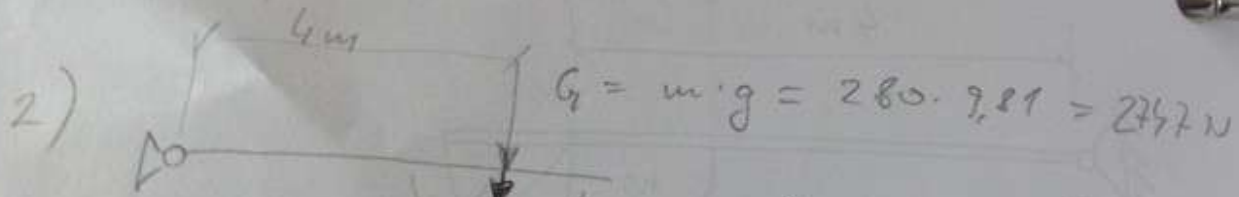
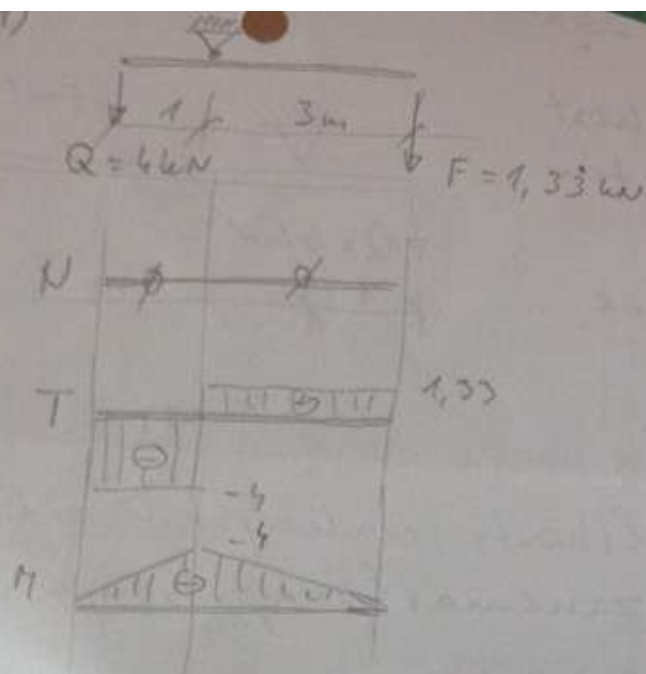
$$r = 1,2 \text{ m}$$

$$m = 280 \text{ kg}$$

strop,
 Elastiko
 ilo 10 N
 stropom
 čunajte

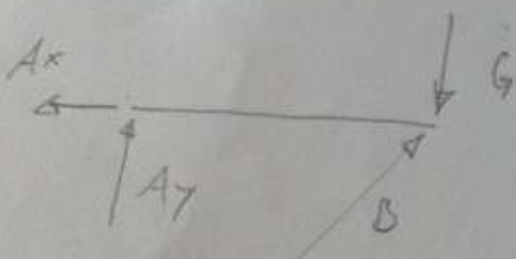
Izračun

1/mr



$$\cos \varphi = \frac{r}{2r} = \frac{1}{2}$$

$$\varphi = 60^\circ$$



$$-A_x + B \cos 60^\circ = 0$$

$$A_y + B \sin 60^\circ - G = 0$$

$$A: -G \cdot 4 + B \sin 60^\circ \cdot 4 = 0$$

$$B = \frac{G}{\sin 60^\circ} = 3172 \text{ N}$$

$$A_x = 1586 \text{ N}$$

$$A_y = 0$$

Izpit iz Mehanike - Statika

OGR, 27.5.2004, 23.1.2006

1) Reducirajte sistem treh sil

$$\vec{F}_1 = (10, -15, 0) \text{ kN} \quad \vec{r}_1 = (2, -5, 0) \text{ m}$$

$$\vec{F}_2 = (5, -8, 0) \text{ kN} \quad \vec{r}_2 = (-1, -2, 0) \text{ m}$$

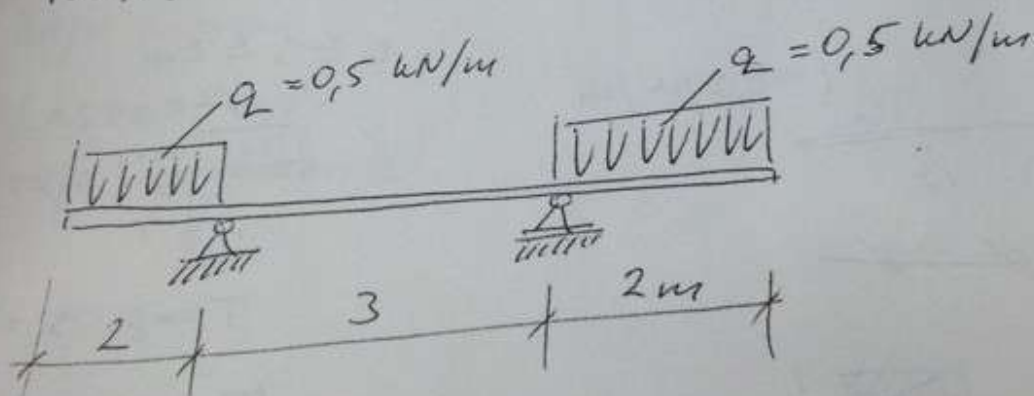
$$\vec{F}_3 = (25, 10, 0) \text{ kN} \quad \vec{r}_3 = (0, -3, 0) \text{ m}$$

a) v točko $T(6, -2, 0) \text{ m}$!

b) v točko $T(0, 0, 0) \text{ m}$!

En konec nenapete elastike privežemo na strop, drugega pa na tla, navpično pod pravim. Elastiko nato potegnemo v vodoravni smeri s silo 10 N tako, da zgornji del elastike odlepa s stropom kot 30° , spodnji pa s tlom 60° . Izračunajte sili v obeh delih elastike!

3) Narišite diagrame N, T, M ! Izračunajte M_{maks} !



$$\begin{aligned} 1) \vec{F}_1 &= (10, -15, 0) \text{ kN} \\ \vec{F}_2 &= (5, -8, 0) \text{ kN} \\ \vec{F}_3 &= (25, 10, 0) \text{ kN} \\ \vec{R} &= (40, -13, 0) \text{ kN} \end{aligned}$$

$$\begin{aligned} \vec{r}_1 &= (2, -5, 0) \text{ m} \\ \vec{r}_2 &= (-1, -2, 0) \text{ m} \\ \vec{r}_3 &= (0, -3, 0) \text{ m} \end{aligned}$$

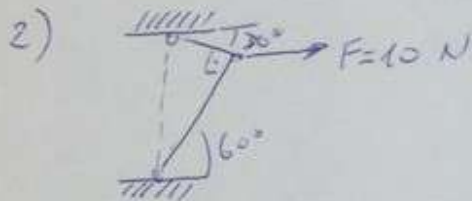
$$\begin{aligned} \vec{M}_1 &= (0, 0, 20) \\ \vec{M}_2 &= (0, 0, 18) \\ \vec{M}_3 &= (0, 0, 75) \end{aligned}$$

$$(y) \vec{M}_0 = (0, 0, 113) \text{ kNm}$$

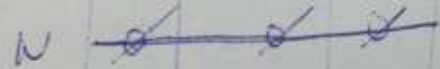
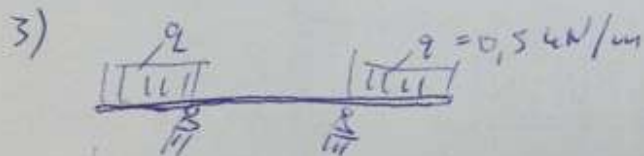
$$\begin{aligned} \vec{d} &= (6, -2, 0) \text{ m} \\ \vec{r}' &= \vec{r} - \vec{d} \end{aligned}$$

$$\begin{aligned} \vec{r}'_1 &= (-4, -3, 0) \text{ m} \\ \vec{r}'_2 &= (-7, 0, 0) \text{ m} \\ \vec{r}'_3 &= (-6, -1, 0) \text{ m} \end{aligned}$$

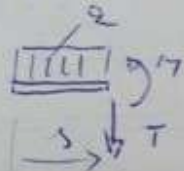
$$(a) \vec{M}_0 = (0, 0, 111) \text{ kNm}$$



$$\begin{aligned} x: F \cdot \cos 60^\circ - F_1 &= 0 \\ y: -F \cdot \sin 60^\circ + F_2 &= 0 \\ F_1 &= 5 \text{ N} \\ F_2 &= 8.66 \text{ N} \end{aligned}$$



$$0 \leq s \leq 2 \text{ m}$$



$$T = -q \cdot s = -1 \text{ kN}$$

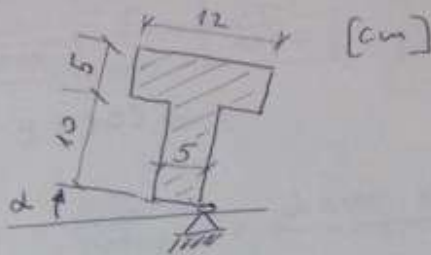
$$M = -q \cdot \frac{s^2}{2}$$

$$M(2) = 1 \text{ kNm}$$

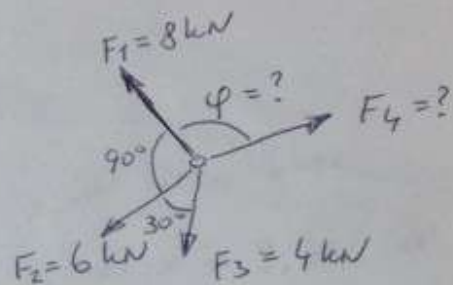
Mehanika - statika

OG, OGR, OMM, 28.6.2006

- 1) Izračunajte, kolikšen nagib α je potreben za prevrnitev predmeta, izrezanega v obliki črke "T"

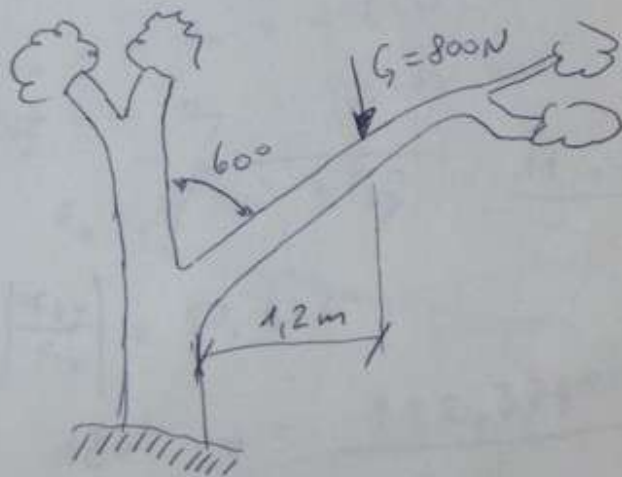


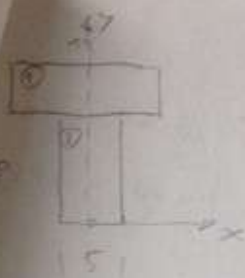
- 2) S kolikšno jakostjo ter pod kakšnim kotom φ mora delovati sila F_4 , da bo doseženo ravnotežje?



- 3) $N, T, M = ?$

Narišite N, T, M za vejo drevesa!
Lastno težo veje zanemarite!





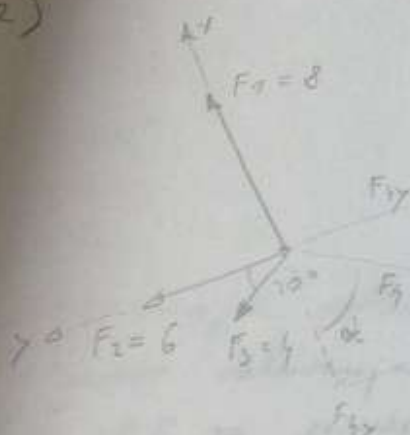
i	A _i	y _i	A _i · y _i
1	60	27,5	1650
2	50	5	250
Σ	110		1900

$$y_T = 9,0909 \text{ cm}$$



$$\tan \alpha = \frac{27,5}{9,0909} \quad ; \quad \alpha = 71,326^\circ$$

2)



$$R_x = F_1 - F_3 \cdot \sin 30^\circ = 6 \text{ kN}$$

$$R_y = F_2 + F_3 \cdot \cos 30^\circ = 9,464 \text{ kN}$$

$$F_{4x} = -R_x = -6 \text{ kN}$$

$$F_{4y} = -R_y = -9,464 \text{ kN}$$

$$F_4 = \sqrt{F_{4x}^2 + F_{4y}^2} = 11,206 \text{ kN}$$

$$\tan \alpha = \left| \frac{F_{4y}}{F_{4x}} \right| = 57,626^\circ$$

$$\varphi = 180^\circ - \alpha = 122,374^\circ$$

$$m, 651 = (1 = 45 \text{ mm})$$

$$m, 651 = (1 = 45 \text{ mm})$$

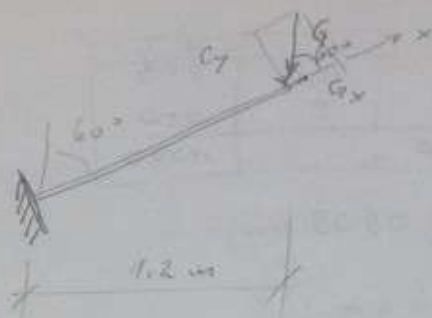
$$m, 651 = (1 = 45 \text{ mm})$$

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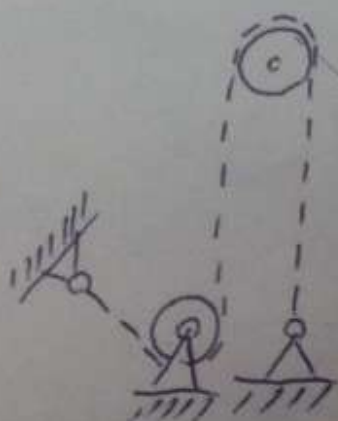
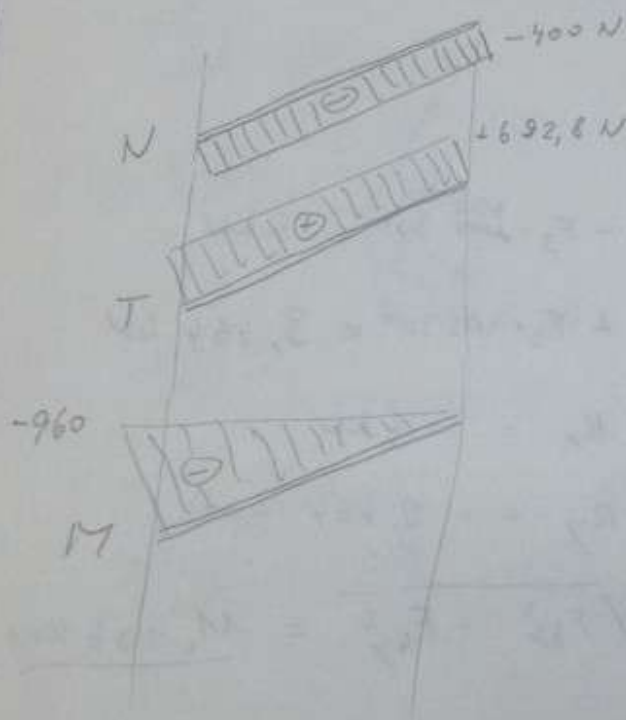
$$m, 651 = (1 = 45 \text{ mm})$$

3)



$$G_x = G \cdot \cos 60^\circ = 400 \text{ N}$$

$$G_y = G \cdot \sin 60^\circ = 692,82 \text{ N}$$

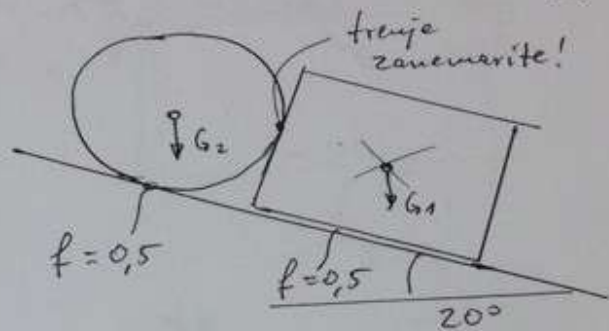


1) Izračunajte, za koliko se spusti točka A, ko sistem obremenimo z utežjo $G = 12 \text{ kN}$!

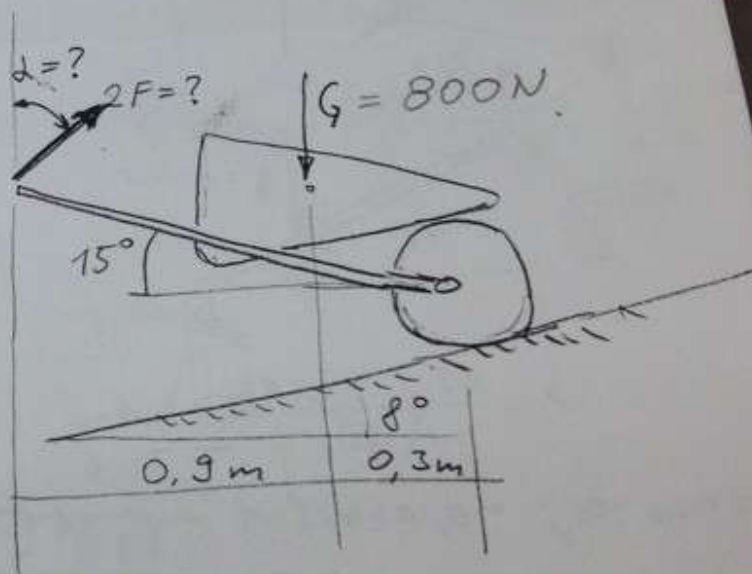
Mehanika - statika

OG - 22.8.2008

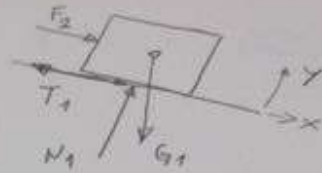
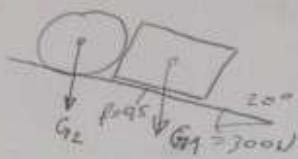
4) Pri kateri teži valja G_2 se sistem začne gibati? $G_1 = 300 \text{ N}$



2) Izračunajte velikost in smer delovanja sile $2F$ za primer mirovanja samokolnice na klanec! Trenje zanemarite.



1)



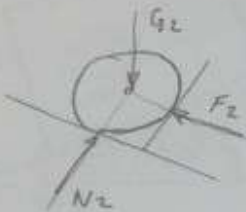
$$\sum F_{ix} = 0; \quad -T_1 + F_2 + G_1 \cdot \sin 20^\circ = 0$$

$$\sum F_{iy} = 0; \quad -G_1 \cdot \cos 20^\circ + N_1 = 0$$

$$T_1 = \mu \cdot N_1$$

$$N_1 = 281,908 \text{ N}, \quad T_1 = 140,91 \text{ N}$$

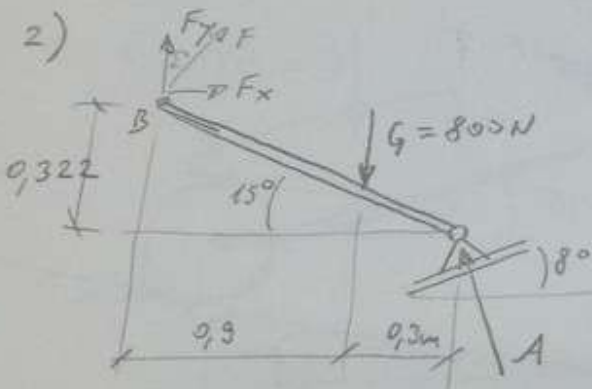
$$F_2 = 38,35 \text{ N}$$



$$\sin 20^\circ = \frac{F_2}{G_2}$$

$$G_2 = F_2 / \sin 20^\circ = 112,12 \text{ N}$$

2)



$$\sum F_{ix} = 0; \quad F_x - A \cdot \sin 8^\circ = 0$$

$$\sum F_{iy} = 0; \quad F_y - G + A \cdot \cos 8^\circ = 0$$

$$\sum M_{iA} = 0; \quad G \cdot 0,3 - F_y \cdot 1,2 - F_x \cdot 0,322 = 0$$

$$\sum M_{iB} = 0; \quad -G \cdot 0,9 + A \cdot \cos 8^\circ \cdot 1,2 - A \cdot \sin 8^\circ \cdot 0,322 = 0$$

$$\sum M_{iB} = 0; \quad -0,04475 \cdot A + 1,1883 \cdot A = G \cdot 0,9$$

$$A \cdot 1,1436 = 720; \quad A = 629,6 \text{ N}$$

$$F_x = 87,62 \text{ N}$$

$$F_y = 176,52 \text{ N}$$

$$2F = \sqrt{F_x^2 + F_y^2} = 197,11 \text{ N}$$

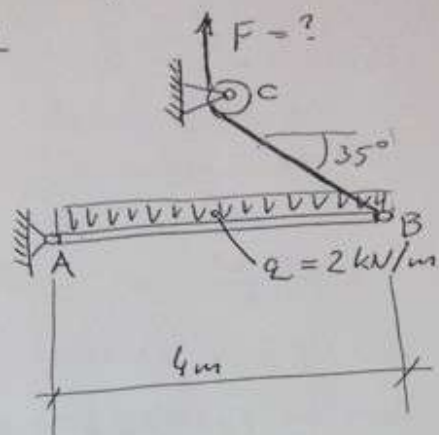
$$\varphi = 26,4^\circ$$

Mechanika - Teil 1

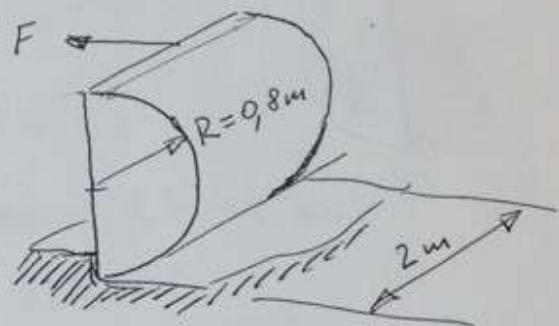
Mehanika - Statika, 12.6.2006

06, 0MM, 06, R

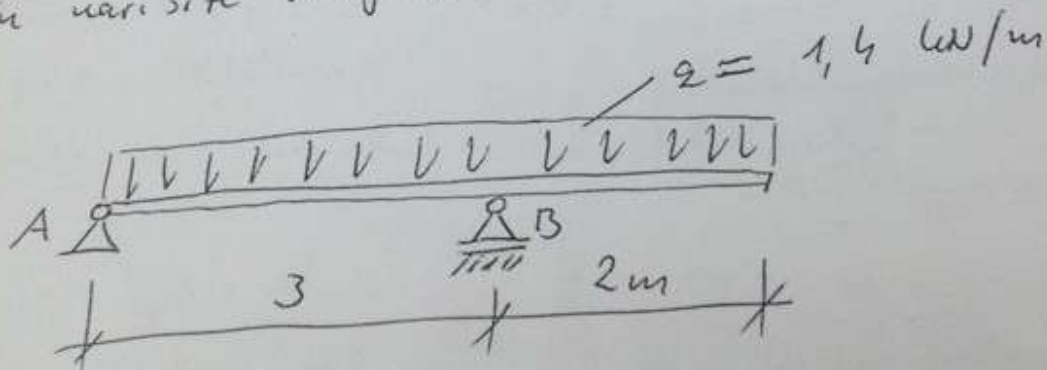
- 1) Izračunajte jakost sile F tako, da bo palica AB stala vodoravno!
Trenja ni!



- 2) S kolikšno silo F je treba delovati na kamniti pol-valj, da bo le-ta ostal v naravnem legu?
 $\rho = 2200 \text{ kg/m}^3$



- 3) Izračunajte N, T, M in narišite diagrame!



1)



$$Q = q \cdot l = 2 \cdot 4 = 8 \text{ kN}$$

$$\sum M_A = 0: -Q \cdot 2 + F \cdot \sin 35^\circ \cdot 4 = 0$$

$$F = \frac{Q \cdot 2}{4 \cdot \sin 35^\circ} = 6,976 \text{ kN}$$

$$2) \quad V = \frac{\pi R^2}{2} \cdot l = \frac{\pi \cdot 0,8^2 \cdot 2}{2} = 2,0106 \text{ m}^3$$

$$m = \rho \cdot V = 4323,3 \text{ kg}$$

$$Q = m \cdot g = 43393 \text{ N}$$



$$e = r_0 = \frac{2 \cdot R \cdot \sin \alpha}{\pi} = \frac{2 \cdot 0,8 \cdot \sin 90^\circ}{\pi/2}$$

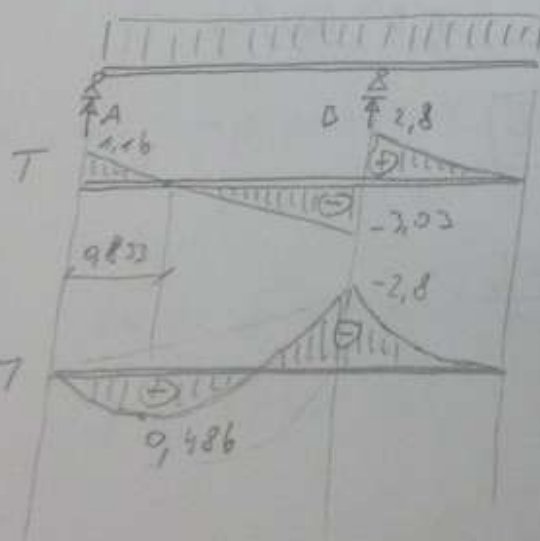
$$e = 2,0106 \text{ m} = 0,33953 \text{ m}$$

$$\sum M_A = 0: F \cdot 2R - Q \cdot e = 0$$

$$F = \frac{Q \cdot e}{2R} = \frac{43393 \cdot 0,3395}{1,6}$$

$$F = 9208 \text{ N}$$

3)



$$\sum M_A = 0: B \cdot 3 - q \cdot 5 \cdot 2,5 = 0$$

$$B = \frac{1,16 \cdot 5 \cdot 2,5}{3} = 5,83 \text{ kN}$$

$$A = q \cdot 5 - B = 1,16 \text{ kN}$$



$$\sum M = 0: A \cdot s - q \cdot \frac{s^2}{2} = 0$$

$$M = -\frac{q}{2} s^2 + A \cdot s$$

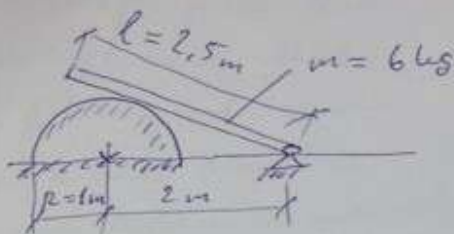
$$\frac{dM}{ds} = -q \cdot s + A = 0$$

$$s = \frac{A}{q} = 0,433 \text{ m}$$

Izpit iz mehanike - statika
2.6.2023 - VS

1)

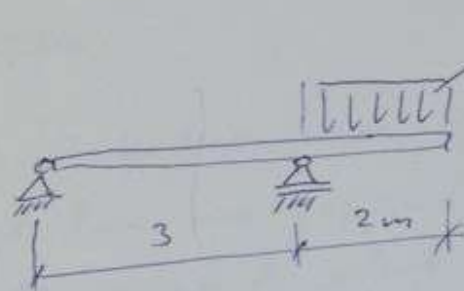
(30)



Izračunajte
velikosti reakcij

2)

(55)

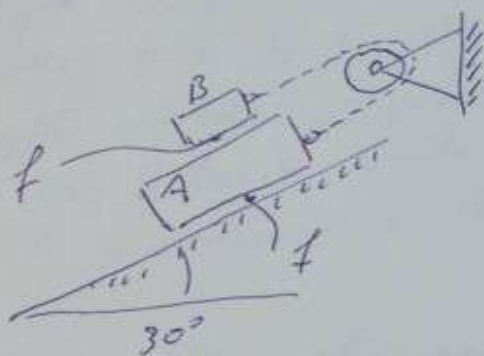


$$q = 4 \text{ kN/m}$$

$N, T, M = ?$

3)

(25)



$$A = 1,5 \text{ kN}$$

$$f = 0,1$$

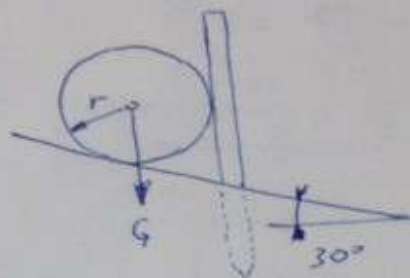
Izračunajte težo ~~A~~ B
tako, da bo A še
miroval in ne bo
zdrsnil navzgor po
hlancu!

Izpit iz mehanike

Statika

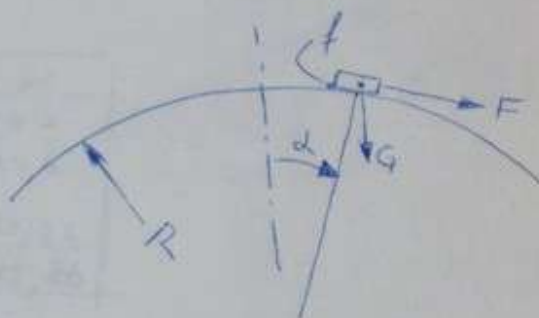
10.4.2006

- 1) Navpično zapičen kol zadržuje valj teže $G = 8 \text{ kN}$ pred zdrsom (treenje ni). Narišite diagram upogibnih momentov, izračunajte M_{maks} !

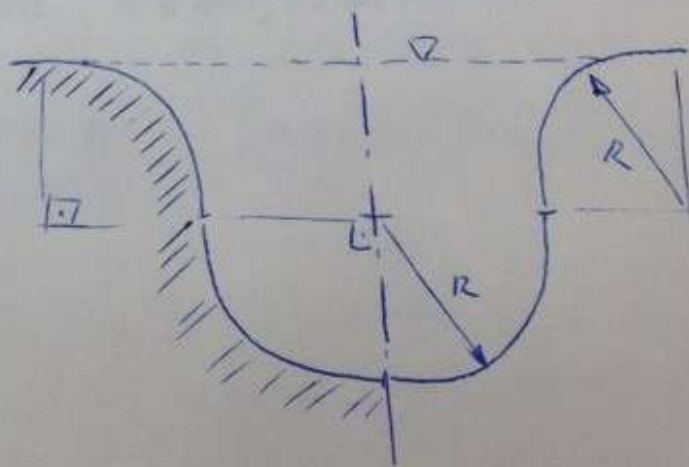


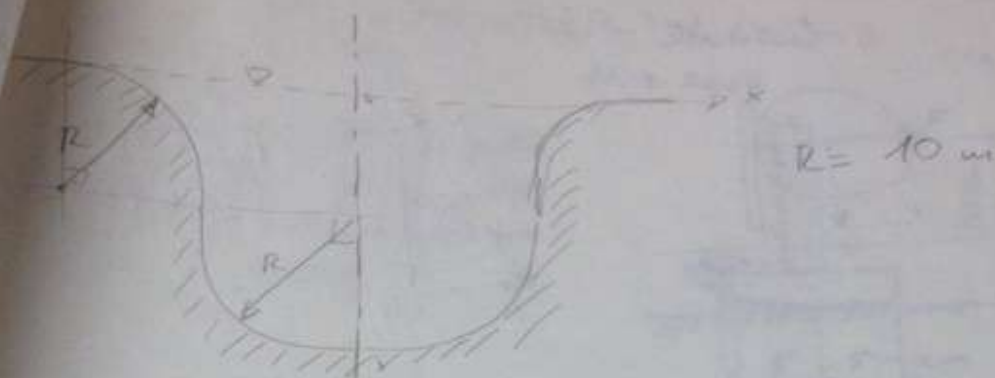
ost

- 2) Sila F , katere smer je tangenta na površino krogle radija R , vleče kvader teže G z vrha krogle navzdol po pobočju ($f = 0,5$). Določite velikost sile F v odvisnosti od kota α ! Pri katerem α je $F = 0$?



- 3) Izračunajte volumen osnosimetrične kotanje, katere prerez je opisan s štirimi krožnimi loki $R = 10 \text{ m}$!





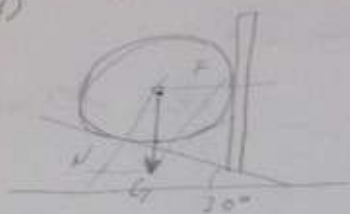
i	A_i	x_i	$A_i \cdot x_i$
1	200	10	2000
2	-78,512	15,756	-1237,6
3	+78,533	4,244	333,32
Σ	200		1095,86

$$y_0 = \frac{2}{3} \cdot R \cdot \frac{\sin 45^\circ}{2} = \frac{2}{3} \cdot 10 \cdot \frac{\sin 45^\circ}{\pi/4} = 6,002 \text{ m}$$

$$x_T = \frac{1095,86}{200} = 5,4793 \text{ m}$$

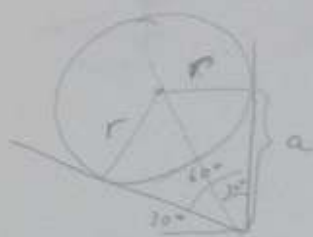
$$V = 2 \cdot \pi \cdot 5,4793 \cdot 200 = 6885,49 \text{ m}^3$$

1) $G = 8 \text{ kN}$, $r = 0,3 \text{ m}$

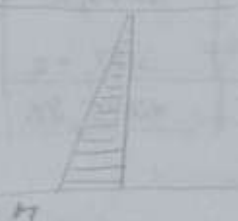
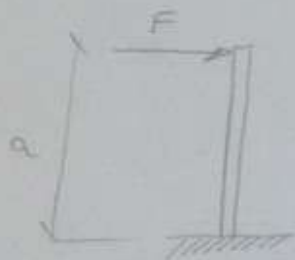


$$F = G \cdot \tan 30^\circ$$

$$F = 4,6188 \text{ kN}$$

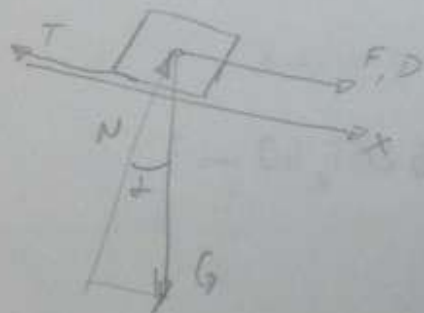


$$\tan 30^\circ = \frac{r}{a}; \quad a = \frac{0,3}{\tan 30^\circ} = 0,5196 \text{ m}$$



$$M = F \cdot a = 2,4 \text{ kNm}$$

2)



$$\sum F_x = 0; \quad F + D - T = 0$$

$$D = G \cdot \sin \alpha$$

$$N = G \cdot \cos \alpha$$

$$T = f \cdot N$$

$$F = T - D = f \cdot N - D = f \cdot G \cdot \cos \alpha - G \cdot \sin \alpha$$

$$F = G (f \cos \alpha - \sin \alpha)$$

$$F = 0 \quad \text{so} \quad f \cos \alpha - \sin \alpha = 0$$

$$f = \frac{\sin \alpha}{\cos \alpha} = \tan \alpha$$

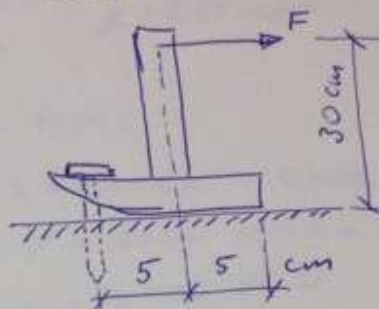
$$\alpha = \arctan 0,5 = 26,56^\circ$$

Izpit iz Mehanike - Statika

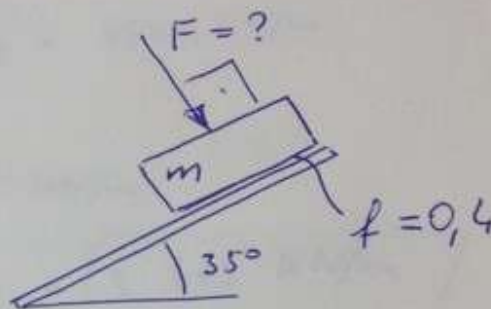
OG

11.4.2007

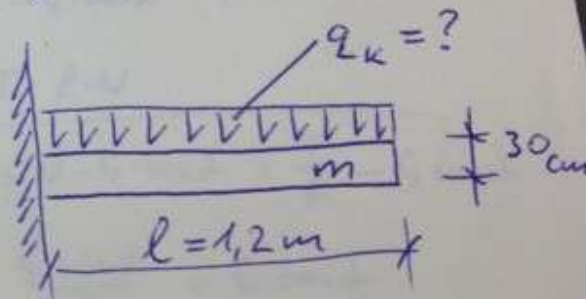
- 1) S kolikšno izvlečno silo delujemo na žebelj, če ročico kladiva vlečemo s silo $F = 180\text{ N}$?



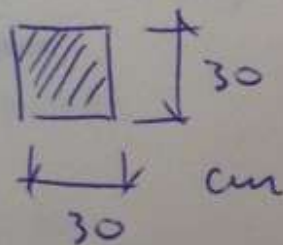
- 2) Delavec po deski z naklonom 35° spušča opetko mase 3 kg . Z najmanj kolikšno silo F mora delovati pravokotno na zgornjo ploskev opetke, da le-ta ne zdrsne? Narišite mnogokotnik sil!

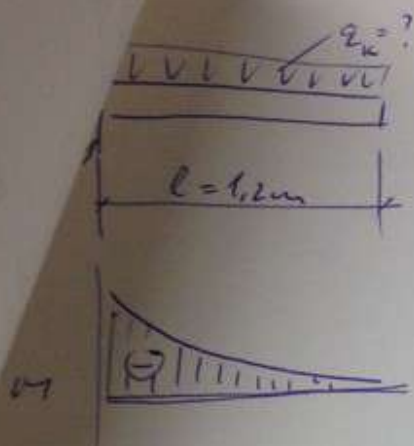


- 3) Kamnita konzola ($\rho = 2200\text{ kg/m}^3$) je poleg lastne teže obremenjena še s "koristno" obtežbo q_k . Kolikšna sme biti ta obtežba, da ekstremni upogibni moment v konzoli doseže absolutno velikost 4 kNm ?



prerez konzole





30
+ +
□ 30 cm

$$\rho = 2200 \text{ kg/m}^3$$

$$g = 9,81 \text{ m/s}^2$$

Lastna težina:

$$V = 0,3 \times 0,3 \times 1,2 = 0,108 \text{ m}^3$$

$$m = \rho \cdot V = 2200 \cdot 0,108 = 237,6 \text{ kg}$$

$$G = m \cdot g = 2330,86 \text{ N}$$

$$q_{LT} = \frac{G}{l} = 1942,4 \text{ N/m}$$

$$M_{maks} = \frac{q \cdot l \cdot l}{2} = 4 \text{ kNm}$$

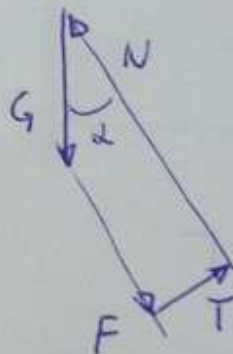
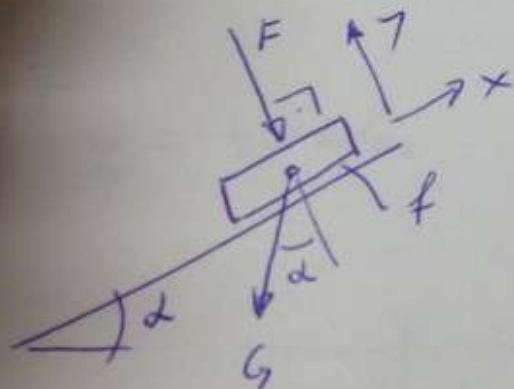
$$q = \frac{2M_{maks}}{l^2} = \frac{2 \cdot 4}{1,2^2} = 5,5 \text{ kN/m}$$

$$q_k = q - q_{LT} = 5,5 - 1,9424 = 3,613 \text{ kN/m}$$

2) $m = 3 \text{ kg}$

$f = 0,4$

$\alpha = 35^\circ$



$$G = m \cdot g = 3 \cdot 9,81 = 29,43 \text{ N}$$

$$x: -G \cdot \sin \alpha + T = 0$$

$$y: -G \cdot \cos \alpha - F + N = 0$$

$$T = f \cdot N$$

$$y: F = N - G \cdot \cos \alpha = \frac{T}{f} - G \cos \alpha$$

$$F = \frac{G \sin \alpha}{f} - G \cos \alpha$$

$$F = G \left(\frac{\sin \alpha}{f} - \cos \alpha \right)$$

$$F = 29,43 \cdot 0,61479$$

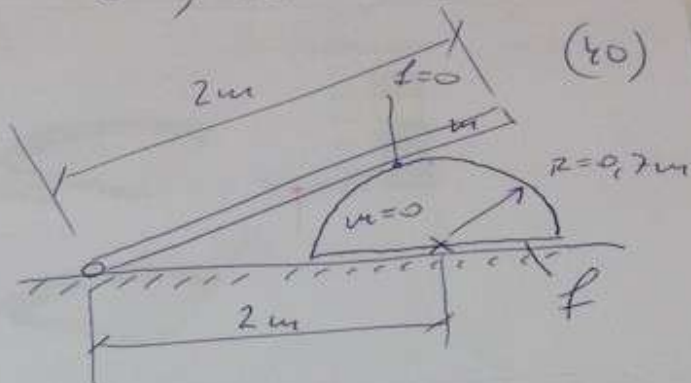
$$F = 18,093 \text{ N}$$

Ispit iz mehanike - statika 1
OGIZ - UN, 22.5.2013

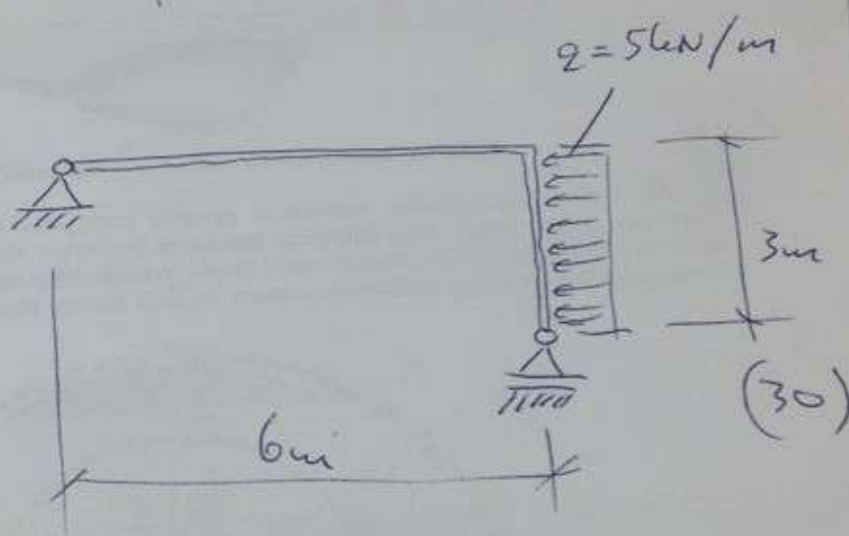
1) $m = 150 \text{ kg}$
 $g = 10 \text{ m/s}^2$

a) Reakcije na
dug

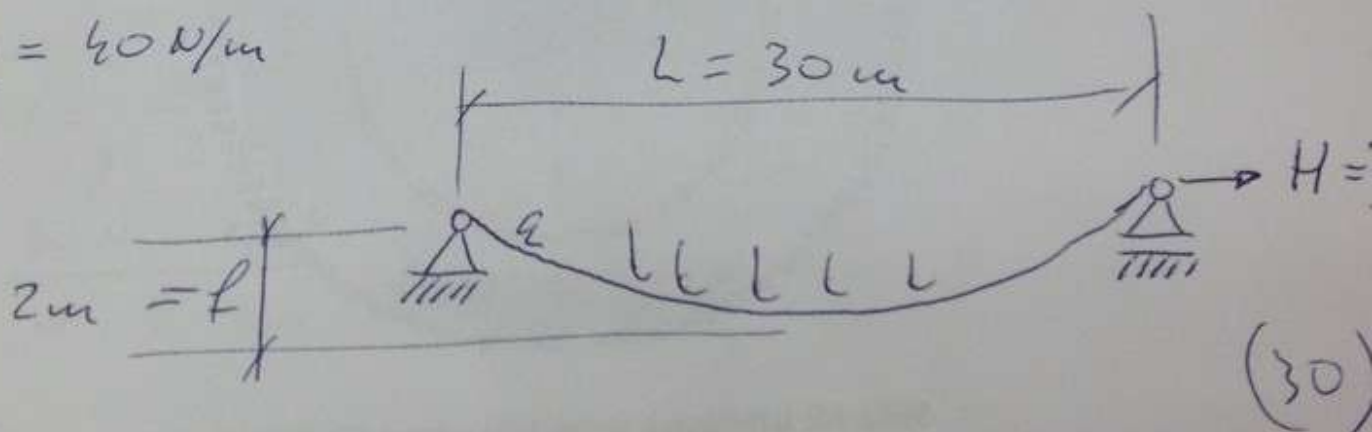
b) $f = ?$



2) $N, T, M = ?$



3) $q = 40 \text{ N/m}$



... v mehaniki

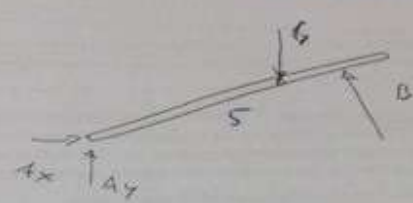
... prebijanja (D=10mm)

... določena oblika odseka

... upoštevavanja ortotropnih

...lipse je velikostnega reda

... evine.



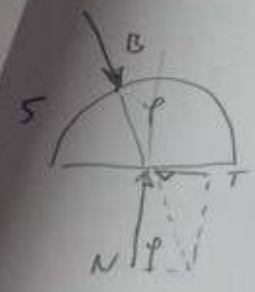
$$150 \text{ kg} \cdot 10 \text{ m/s}^2 = 1500 \text{ N}$$

$$\sum M_i A = 0; 0 = B \cdot 1.8735 - G \cdot 1 \cdot \cos 20.487^\circ$$

$$B = 750 \text{ N}$$

$$= B \cdot \sin \varphi = 262.5 \text{ N}$$

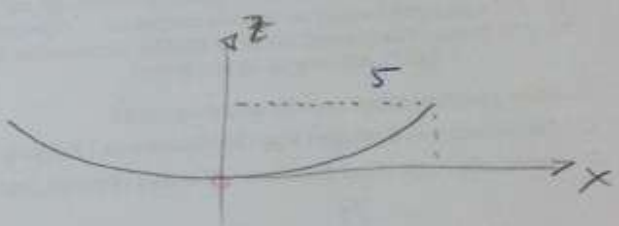
$$= G - B \cdot \cos \varphi = 797.4 \text{ N}$$



$$T = N \cdot f$$

$$f = \frac{T}{N} = \tan \varphi = \underline{\underline{0.3736}}$$

$$z = \frac{q x^2}{2H}$$



$$x = 15 \text{ m}$$

$$z = 2 \text{ m}$$

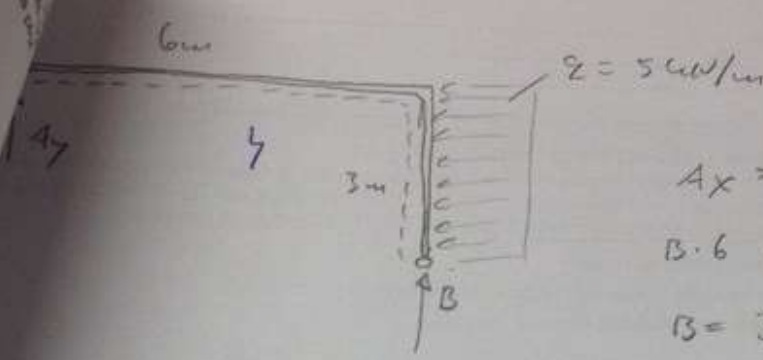
$$q = 40 \text{ N/m}$$

$$H = \frac{q x^2}{2z}$$

$$H = \frac{40 \text{ N} \cdot 15^2 \text{ m}^2}{\text{m} \cdot 2 \cdot 2 \text{ m}}$$

$$H = 2250 \text{ N}$$

mek. lina 111.m
 $q = 15 \text{ kN/m}$
 $L = 5 \text{ m}$
 $E = 2 \cdot 10^5 \text{ MPa}$
 $J = 4 \cdot 10^6 \text{ mm}^4$

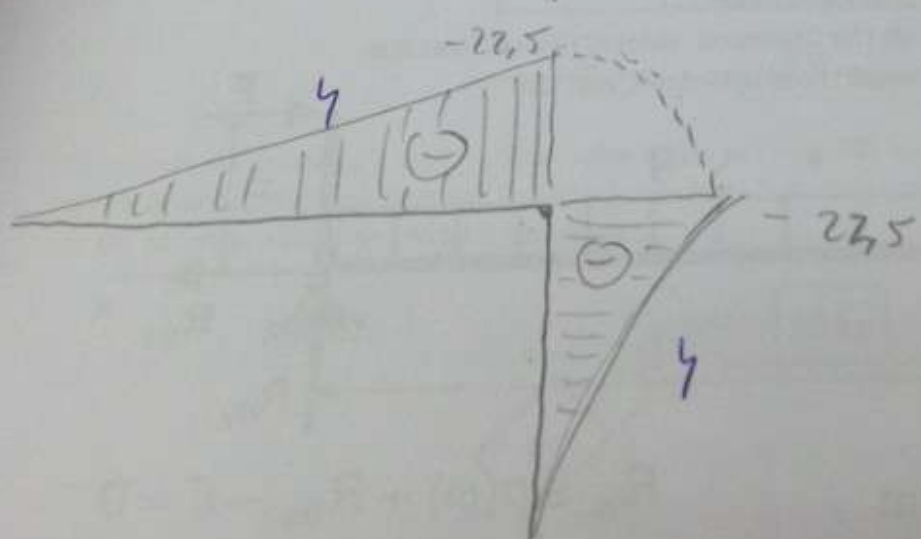
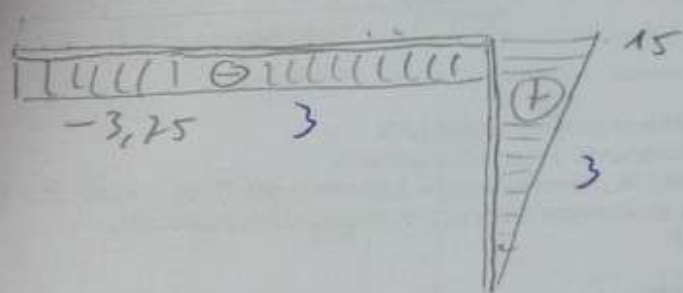
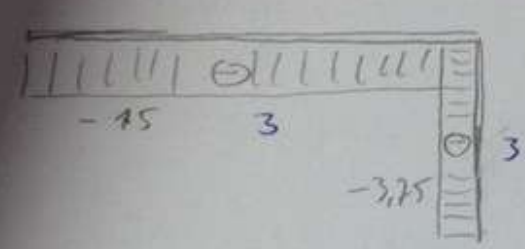


$$A_x = q \cdot 3 = 15 \text{ kN} \quad 2$$

$$B \cdot 6 - q \cdot 3 \cdot 1,5 = 0$$

$$B = 3,75 \text{ kN} \quad 2$$

$$A_y = -3,75 \text{ kN} \quad 2$$

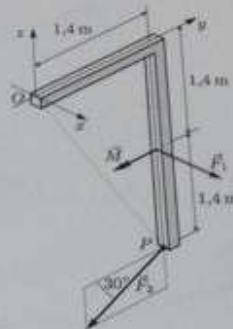


5. IZPIT IZ STATIKE
NTF Geotehnologija in rudarstvo VSS
27. avgust 2012

Ta list je vaš. Sklepe, ki definirajo nastopajoče veličine, sodijo v vaš izdelek. V le-tem naj bo razviden postopek reševanja. Naloge so vredne: prva 35, druga 35 in tretja 30 točk.

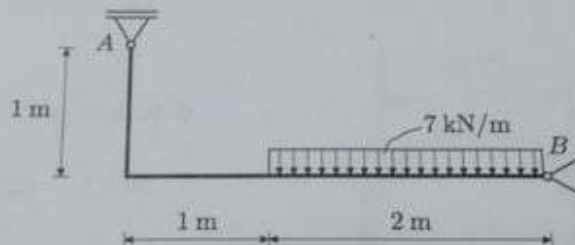
- (1) Za prikazani sistem sil velja: $|\vec{F}_1| = 3 \text{ kN}$, $|\vec{F}_2| = 7 \text{ kN}$, $|\vec{M}| = 5 \text{ kNm}$.

- (a) Izvedite redukcijo prikazanega sistema sil v koordinatno izhodišče O .
(b) Določite, s kolikšnim momentom deluje ta sistem sil okrog osi OP .

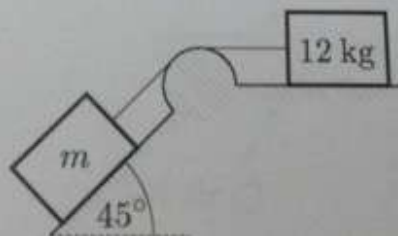


- (2) Za konstrukcijo na sliki:

- (a) Izračunajte reakcije v podporah.
(b) Izrišite diagrame notranje osne sile, prečne sile in notranjega upogibnega momenta.
(c) Določite maksimalni upogibni moment in mesto, na katerem se pojavi.

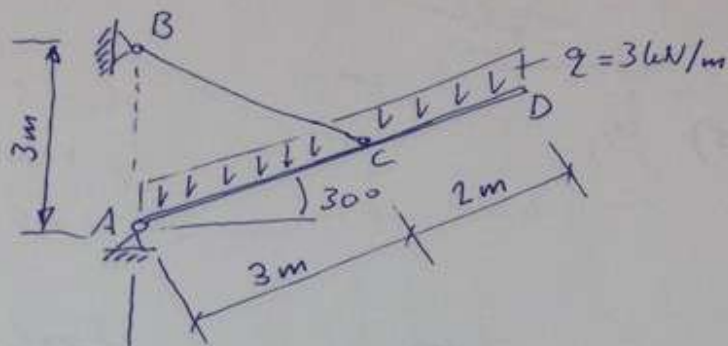


- (3) Dve kladi sta povezani z vrvjo. Masa desne klade je 12 kg. Na vseh kontaktih (tudi v kontaktu vrvi s krožno površino) nastopa trenje s količnikom 0,2. Izračunajte največjo maso leve klade m , ob kateri je sistem še vedno v ravnovesju.

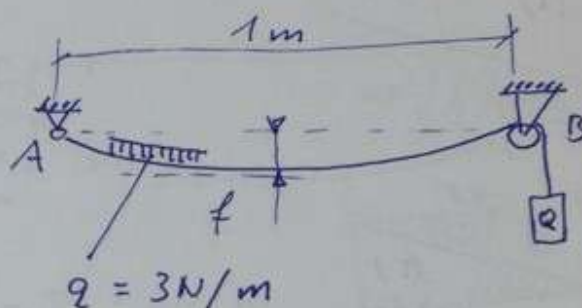


Izpit iz Statike 1, 21. 1. 2013

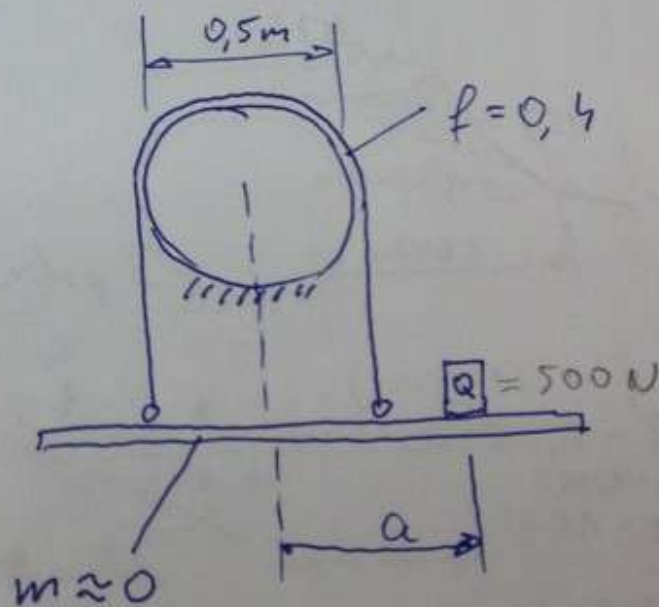
- 1) Izračunajte in narišite diagrame N T M .



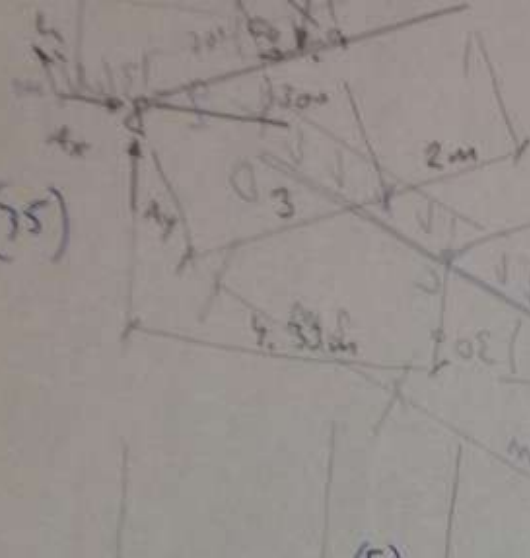
- 2) Določite velikost uteži Q tako, da bo povprečje $f = 0,06m$! Trenja ni.



- 3) Izračunajte največjo razdaljo a tako, da bo sistem še v mirovanju!



(75)



$$\sum M_A = 0; C \cdot \cos 30^\circ \cdot 3 = Q \cdot \frac{3.3}{2}$$

$$C = 10.825 \text{ kN}$$

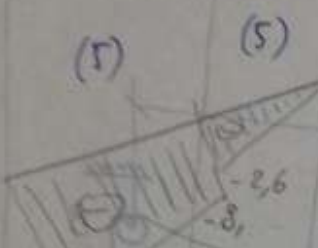
$$x: A_x - C \cdot \cos 30^\circ = 0, A_x = 9.374 \text{ kN}$$

$$A_y + C \cdot \sin 30^\circ - Q = 0$$

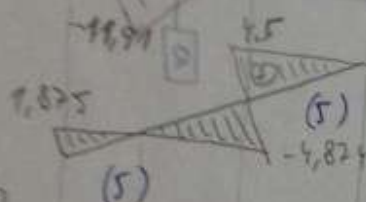
$$A_y = 2.577 \text{ kN}$$

$$I_{pol} \text{ e } 0 \leq s \leq 1.732 \text{ m}$$

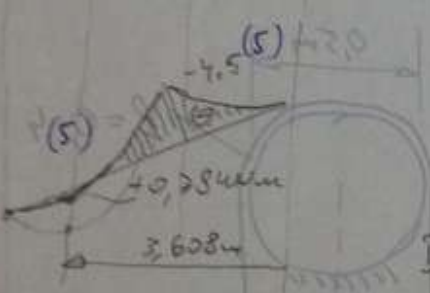
N [kN]



T [kN]



M [kNm]



$$N + 2 \cdot 3 \cdot \sin 30^\circ = 0$$

$$N = -3.05 \text{ kN}$$

$$N(0) = 0$$

$$N(1.732 \text{ m}) = -2.598 \text{ kN}$$

$$m/UE = \sum T - 2 \cdot 3 \cdot \cos 30^\circ = 3.4 \text{ kN}$$

$$T = 2.598 \cdot 3, T(0) = 0$$

$$T(1.732) = 4.5 \text{ kN}$$

$$M + 2 \cdot \frac{3^2}{2} = 0, M = -9 \text{ kNm}$$

$$M(1.732) = -4.5 \text{ kNm}$$

I pol e

$$1.732 \leq s \leq 4.33$$

$$N + 2 \cdot 3 \cdot \sin 30^\circ + C_{II} = 0$$

$$N = -5.425 - 1.5 \cdot 3$$

$$N(1.732) = -5.425$$

$$N(4.33) = -11.91 \text{ kN}$$

$$N(4.33) = -11.91 \text{ kN}$$

$$T - 2 \cdot 3 \cdot \cos 30^\circ + C_{II} = 0$$

$$T(1.732) = 2.598 \cdot 3 - 9.374 = -4.874 \text{ kN}$$

$$T(4.33) = 1.875 \text{ kN}$$

$$M + 2 \cdot \frac{3^2}{2} - C_{II} \cdot 3 = 0$$

$$M = 9.374 \cdot 3 - 1.5 \cdot 3^2$$

$$M(4.33) = -4.5 \text{ kNm}$$

$$M(4.33) =$$

2)
(30)



$$z = \frac{g}{2H} \cdot x^2$$

A dan B : $x = 0.5 \text{ m}$
 $z = 0.06 \text{ m}$

(10) $V = g \cdot x = 0.5 \cdot 3 = 1.5 \text{ N}$

$$H = \frac{g \cdot x^2}{2 \cdot z} = \frac{3 \text{ N} \cdot 0.5^2 \text{ m}^2}{\text{m} \cdot 2 \cdot 0.06 \text{ m}}$$

(10) $Q = \sqrt{V^2 + H^2}$
 $Q = 6.427 \text{ N}$

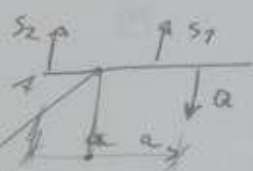
$H = 6.25 \text{ N}$ (10)

3)

(35)



$S_1 > S_2$; $S_1 = S_2 \cdot e^{\mu \theta}$; $e^{\mu \theta} = e^{0.4 \cdot \pi} = 2.13$
 (10) $S_1 = 3.513 S_2$; $S_1 = 389.2 \text{ N}$



(10) $\downarrow S_1 + S_2 = Q$; $3.513 S_2 + S_2 = Q$; $S_2 = \frac{Q}{4.513}$
 $S_2 = 110.8 \text{ N}$

$$\sum M_{iA} = 0 \quad - S_1 \cdot 0.5 + Q \cdot a = 0$$

$$a = \frac{Q}{S_1 \cdot 0.5} = \frac{500}{389.2 \cdot 0.5} = 2.569 \text{ m}$$

$\sum M_{iA} = 0$; (10) $Q \cdot a + S_2 \cdot 0.25 - S_1 \cdot 0.25 = 0$

(15) $Q \cdot a + 27.7 - 97.3 = 0$

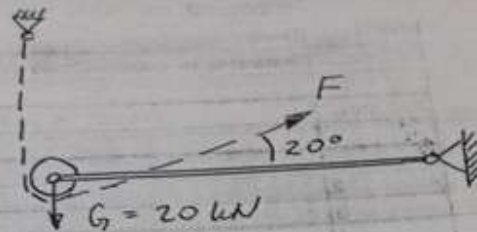
$$a = \frac{69.6}{500} = 0.139 \text{ m}$$

(5)

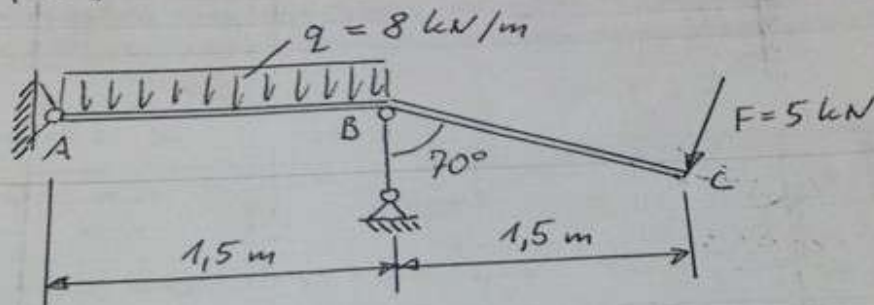
Izpit iz Mehanike, 4. 1. 2007

0177 - 062 - statika

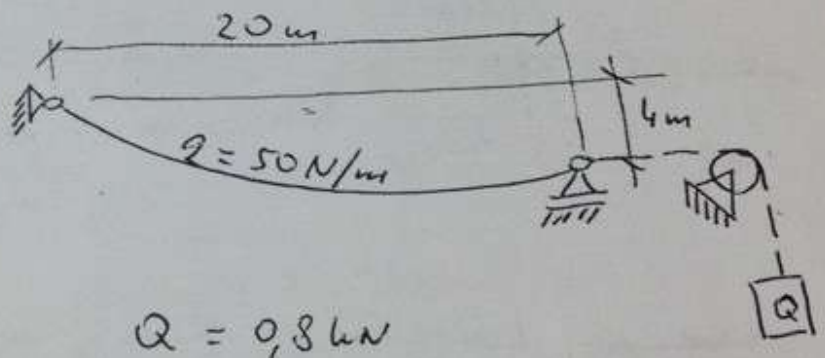
- 1) Določite silo F tako, da bo sistem v ravnotežju! Izmere kolokta zanemarite.



- 2) $N, T, M = ?$



- 3) Izračunajte lego temena vrvi, isto maksimalno silo v vrvi in dolžino vrvi. Predpostavite parabolično verigo.



Izpit iz Mehanike

Statika

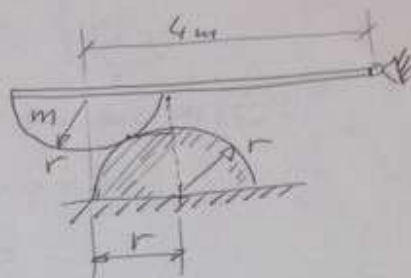
27. 6. 2007

1) Izračunajte velikosti reakcij!

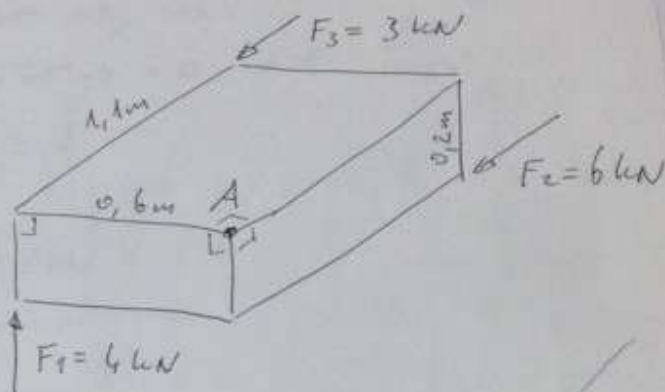
Masa droga je zanemarljiva,
trenja ni.

$$M = 280 \text{ kg}$$

$$r = 1,2 \text{ m}$$



2) Reducirajte sistem
sil v točko A!

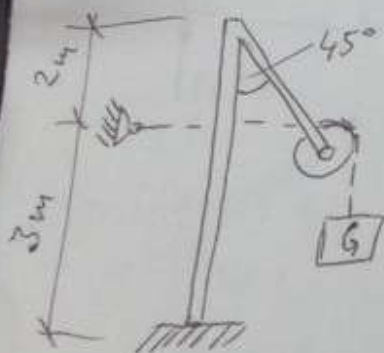


3) OGR, OMM

Narisite N, T, M !

$$G = 2 \text{ kN}$$

Trenja ni!

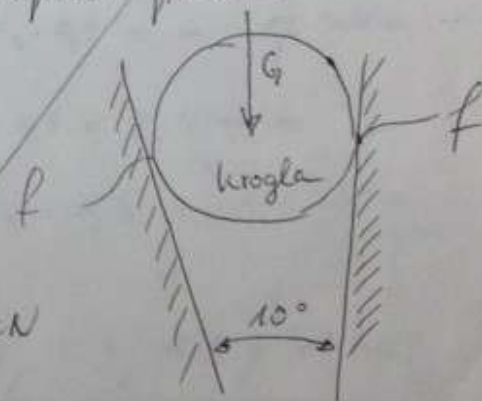


3) OG

Izračunajte velikost dvizne
sile F za dvig bremena G ,
ki se je zagledilo med
dve krupavi ploskvi

$$G = 11 \text{ kN}$$

$$f = 0,6$$



Ker telo ni deform., dobimo
trivredni sistem



$$\varphi = \arccos\left(\frac{r}{2r}\right)$$

$$\varphi = 60^\circ$$

$$G = 280 \cdot 9,81 = 2747 \text{ N}$$

$$\sum F_x = 0; -1 \cdot \cos 60^\circ + B_x = 0$$

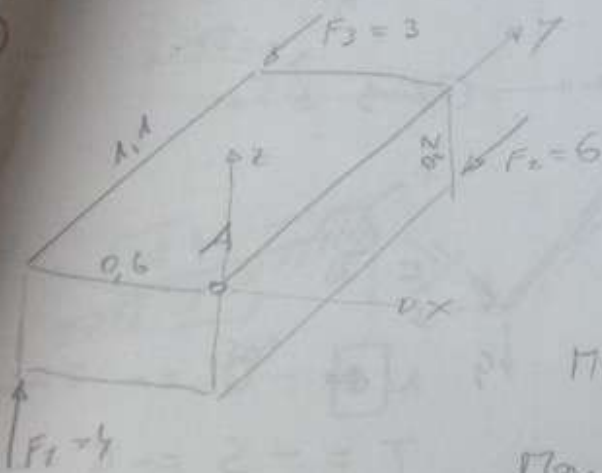
$$\sum F_y = 0; -G + 1 \cdot \sin 60^\circ + B_y = 0$$

$$\sum M_C = 0; G \cdot 4 - 1 \cdot \sin 60^\circ \cdot 4 = 0 \rightarrow 1 = \frac{G}{\sin 60^\circ}$$

$$1 = 3122 \text{ N}$$

$$B_y = 0$$

$$B_x = 1586 \text{ N}$$



$$R_x = 0$$

$$R_y = -6 + 3 = -3 \text{ kN}$$

$$R_z = 4 \text{ kN}$$

$$\vec{R} = (0, -3, 4) \text{ kN}$$

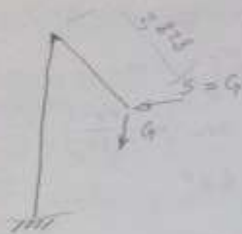
$$M_{Ox} = -F_2 \cdot 0,2 = -1,2 \text{ kNm}$$

$$M_{Oy} = +F_1 \cdot 0,6 = 2,4 \text{ kNm}$$

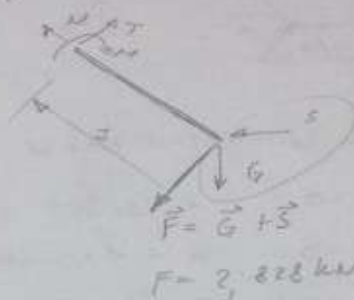
$$M_{Oz} = +F_3 \cdot 0,6 = 1,8 \text{ kNm}$$

$$\vec{M}_O = (-1,2, 2,4, 1,8) \text{ kNm}$$

2) 04R, 0111



I polje: $0 \leq s < 2,828$



$$F = 2,828 \text{ kN}$$

$$N = 0$$

$$T = F = 2,828 \text{ kN}$$

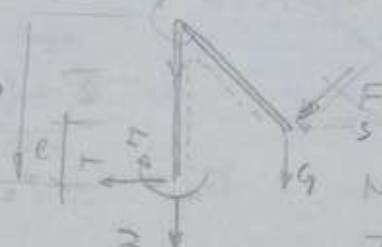
$$M + F \cdot s = 0$$

$$M = -F \cdot s ; M_0 = 0$$

$$M(2,828) = -8 \text{ kNm}$$



II polje: $0 \leq s \leq 5$



$$N = -G = -2 \text{ kN}$$

$$T = -S = -G = -2 \text{ kN}$$

$$M + G \cdot 2 + S \cdot (2 - s) = 0$$

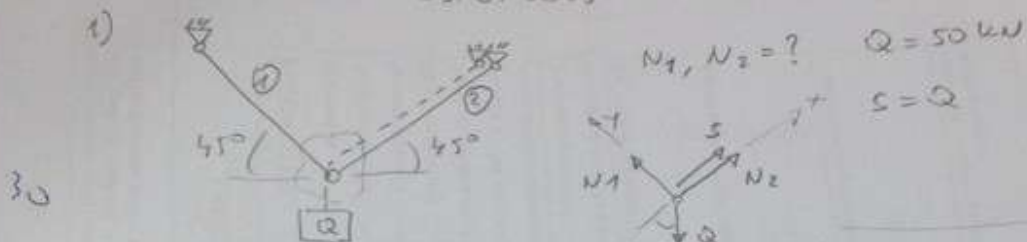
$$M = -4 - 2(2 - s)$$

$$M(0) = -4 - 4 = -8 \text{ kNm}$$

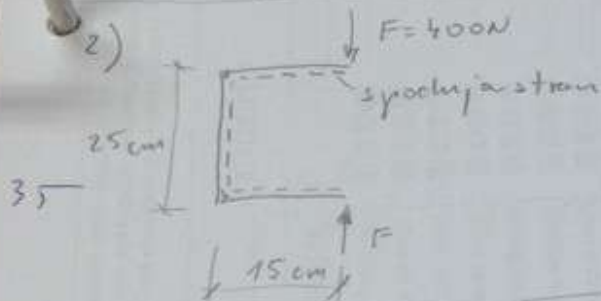
$$M(5) = -4 - 2(-3) = 2 \text{ kNm}$$



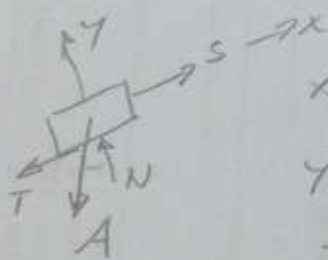
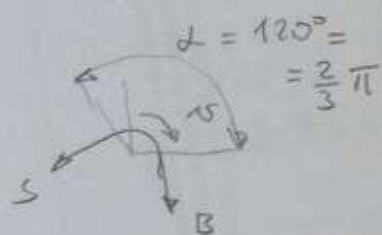
Statika montazisti
23.6.2025



$\sum F_{ix} = 0; N_2 + S_2 - Q \cdot \cos 45^\circ = 0 \Rightarrow N_2 = -14,64 \text{ kN}$
 $\sum F_{iy} = 0; N_1 - Q \cdot \sin 45^\circ = 0 \Rightarrow N_1 = 35,35 \text{ kN}$



$A = 500 \text{ N}$
 $f = 0,4$



$x: S - T - A \cdot \sin 30^\circ = 0$

$y: N - A \cdot \cos 30^\circ = 0$

$T = f \cdot N$

$N = 433,01 \text{ N}$

$T = 173,2 \text{ N}$

$S = 423,2 \text{ N}$

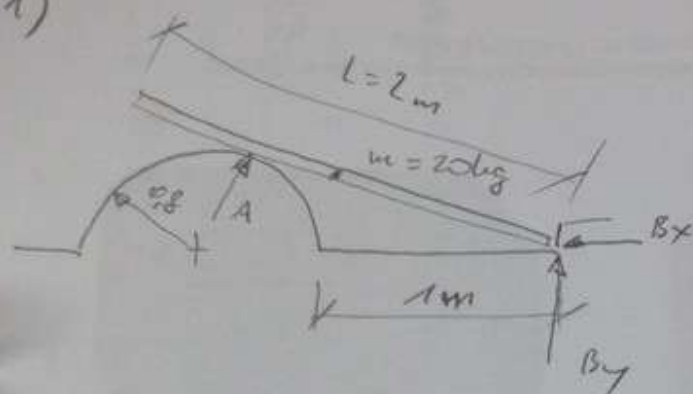
$B = S \cdot e^{f\alpha}$

$B = 423,2 \cdot e^{0,4 \cdot \frac{2}{3}\pi}$

$B = 978,1 \text{ N}$

Izpit iz Statike, NTF, OMT, OGR ①
28.1.2015

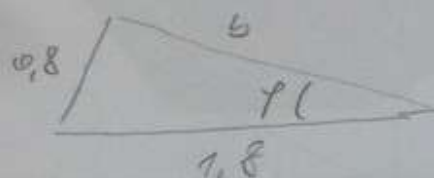
1)



Reakcije:

$$G = m \cdot g = 20 \cdot 9,81$$

$$G = 196,2 \text{ N}$$



$$\sin \varphi = \frac{0,8}{1,8} ; \varphi = 23,39^\circ ;$$

$$b = 1,612 \text{ m}$$

$$\sum F_{ix} = 0 : A \cdot \sin \varphi - B_x = 0$$

$$\sum F_{iy} = 0 : A \cdot \cos \varphi + B_y - G = 0$$

$$\sum M_{iB} = 0 ; G \cdot \frac{L}{2} \cdot \cos \varphi - A \cdot b = 0$$

$$B_x = 44,35 \text{ N}$$

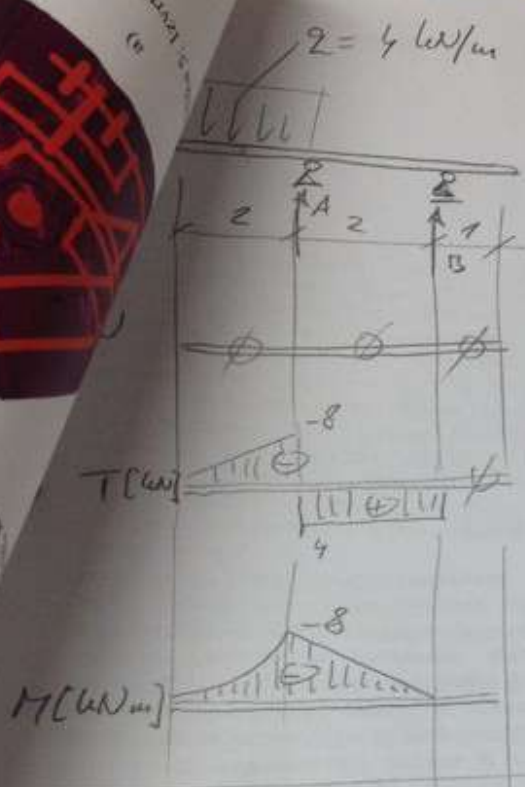
$$B_y = 93,67 \text{ N}$$

$$111,21$$

$$\Rightarrow A = \frac{48,30}{1,612}$$



Slika 4: Gibljiva je ...
... so namenske preloke ...
... element geometrije ...

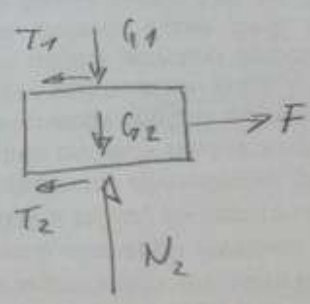
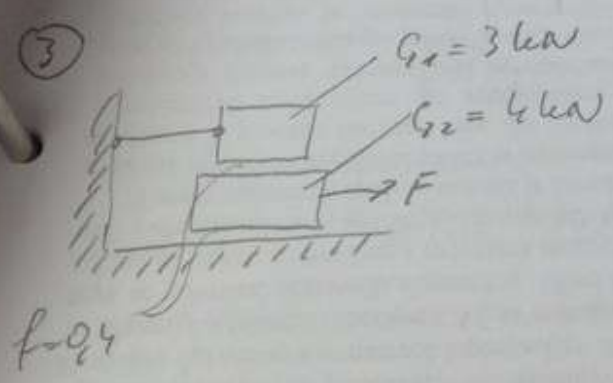


②

$$\sum M_i B = 0 ; 4 \cdot 2 - 2 \cdot 2 \cdot 3 = 0$$

$$A = 12 \text{ kN}$$

$$B = -4 \text{ kN}$$



$$T_1 = G_1 \cdot f = 1.2 \text{ kN}$$

$$N_2 = G_1 + G_2 = 7 \text{ kN}$$

$$T_2 = N_2 \cdot f = 2.8 \text{ kN}$$

$$F = T_1 + T_2 = \boxed{4.0 \text{ kN}}$$

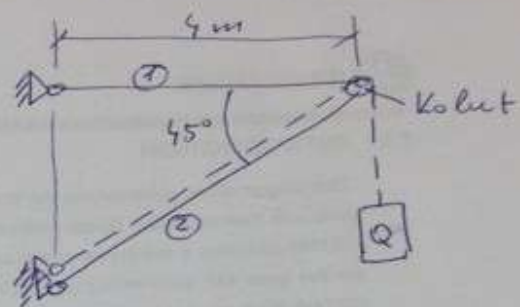
Izpit iz Statike 1, 4.2.2013

1) Izračunajte sile v

palicah 1 in 2!

Trenja ni, kolček je zanemarljivo majhen.

$Q = 14 \text{ kN}$
(25 točk)



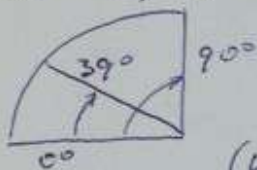
2) Narišite diagrame

N, T, M . Pri krožnem

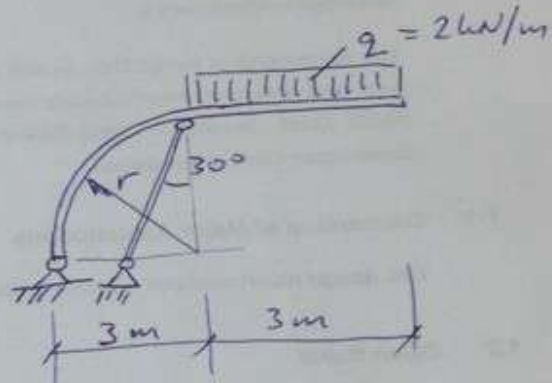
loku izračunajte

N, T in M samo za

koote $0^\circ, 39^\circ$ in 90° .



(45 točk)



3) Z gumijastim kolesom

premera 90 cm želimo

speljati z mesta (brez

zaleta) preko praga višine

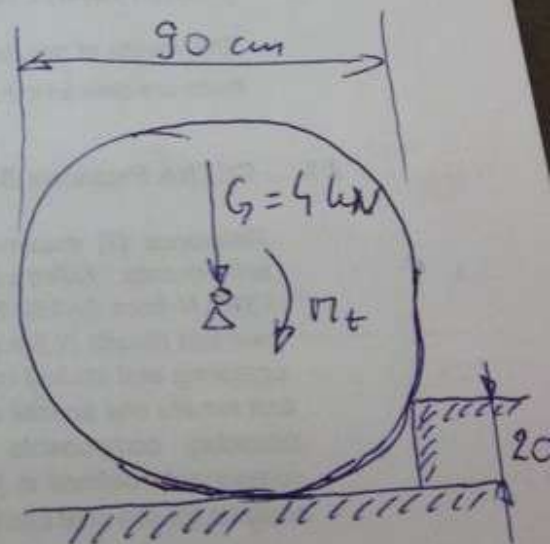
20 cm. Kolikšen koeficient

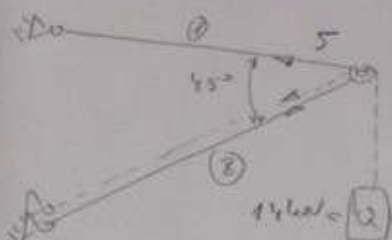
trenja med kolesom in

pragom bi bil za to potreben?

Kolikšen torzijski moment

na osi kolesa potrebujemo? (30 točk)





$$\sum F_x = N_1 \cdot \cos 45^\circ - N_2 \cdot \cos 45^\circ - 20 = 0$$

$$\sum F_y = N_2 \cdot \sin 45^\circ - 5 \cdot \sin 45^\circ - 14 = 0$$

$$S = 2$$

(25)

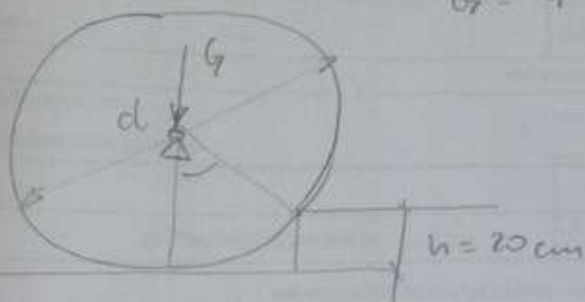
$$\sum N_x = -33,8 \text{ kN} \quad \text{+plate!}$$

$$\sum N_y = 14 \text{ kN}$$

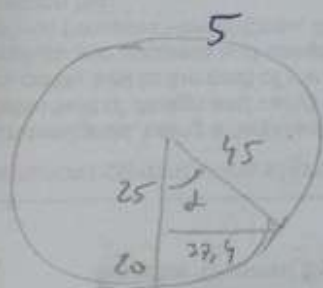
3)

$$d = 90 \text{ cm}$$

$$G = 4 \text{ kN}$$



(30)



$$\cos \alpha = \frac{25}{45}; \quad \alpha = 56,25^\circ$$

$$\alpha = \alpha_0; \quad f = \tan \alpha_0$$

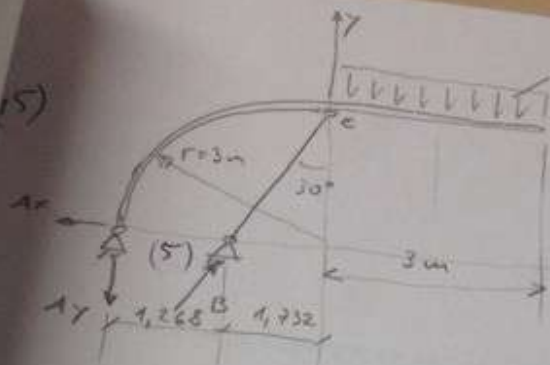
$$f = 1,496 \approx \underline{\underline{1,5}}$$

$$\sum M_t = \frac{r \cdot f \cdot G}{\cos \alpha + f \cdot \sin \alpha} = \frac{0,45 \cdot 1,496 \cdot 4000}{\cos 56,25 + 1,496 \cdot \sin 56,25}$$

$$\sum M_t = \underline{\underline{1496 \text{ Nm}}}$$

ali

$$M_t = G \cdot r \cdot \sin \alpha = G \cdot r_0 = 4000 \cdot 0,374 = 1496 \text{ Nm}$$



$$q = 2 \text{ kN/m}$$

Realizacija:

$$\sum F_{ix} = 0; -A_x + B \cdot \sin 30^\circ = 0$$

$$\sum F_{iy} = 0; -A_y - 2 \cdot 3 + B \cos 30^\circ = 0$$

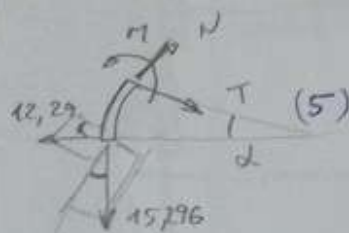
$$\sum M_{iA} = 0; -2 \cdot 3 \cdot 1.5 + B \cos 30^\circ \cdot 1.268 = 0$$

$$B = 24.59 \text{ kN}$$

$$A_x = 12.29 \text{ kN} \quad (5)$$

$$A_y = 15.296 \text{ kN}$$

Univrljeno polje: $0^\circ \leq \varphi \leq 90^\circ$

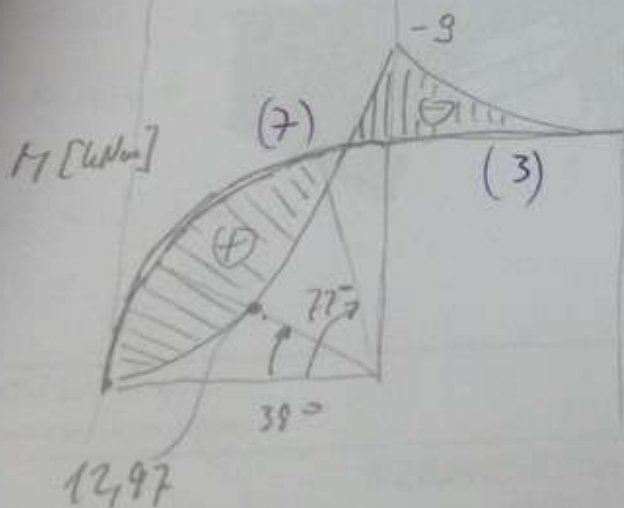
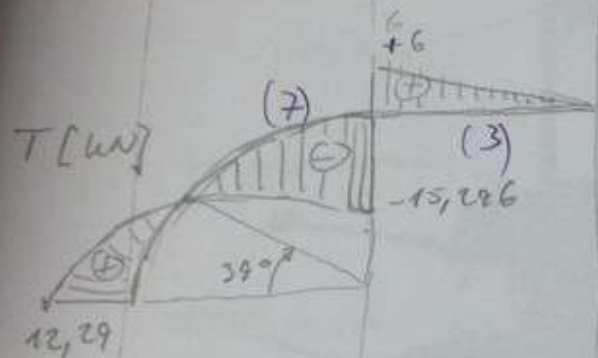
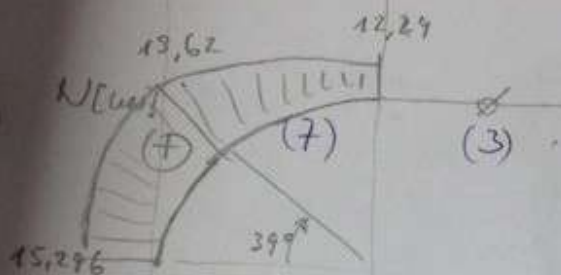


$$N - 12.29 \cdot \sin \varphi - 15.296 \cdot \cos \varphi = 0$$

$$N(0^\circ) = 15.296 \text{ kN}$$

$$T - 12.29 \cdot \cos \varphi + 15.296 \cdot \sin \varphi = 0$$

$$M + 15.296(3 - 3 \cdot \cos \varphi) - 12.29 \cdot 3 \cdot \sin \varphi = 0$$



Izpit iz Statike 1, 7. III 2013

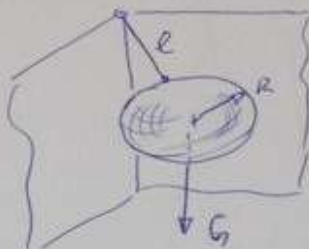
- 1) Izračunajte silo v vrvi in silo sten na kroglo!

Trenja ni.

$$G = 120 \text{ N}$$

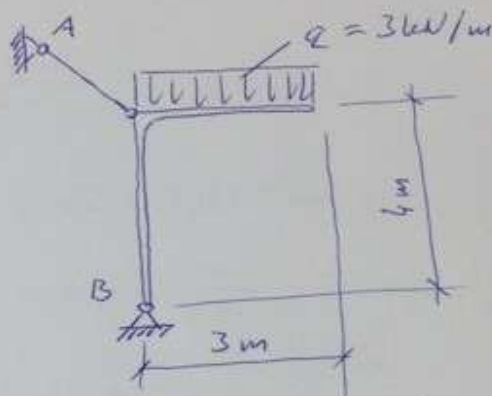
$$R = 6 \text{ cm}$$

$$l = 10 \text{ cm}$$



(35 točk)

- 2) Izračunajte velikosti reakcij, izračunajte in narišite N , T , M !



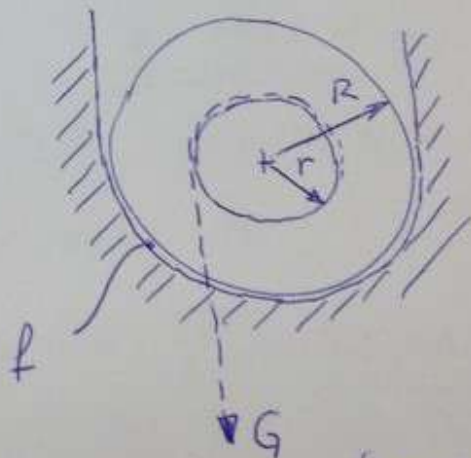
(35 točk)

- 3) Vsaj kolikšen mora biti polmer r , da se kolut R prične obračati?

$$R = 5 \text{ cm}$$

$$G = 200 \text{ N}$$

$$f = 0,5$$



(30 točk)

$$M_t = \frac{R \cdot f \cdot G}{\cos \alpha + f \sin \alpha} \quad (10)$$

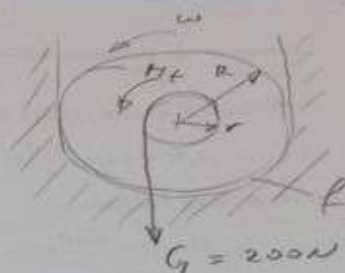
$$f = 0,5; \quad \alpha = \arctg 0,5$$

$$\alpha = 26,565^\circ$$

$$R = 0,05 \text{ m}$$

$$M_t = \frac{0,05 \cdot 0,5 \cdot 200 \text{ N}}{\cos 26,6^\circ + 0,5 \cdot \sin 26,6^\circ} = 4,472 \text{ Nm} \quad (10)$$

$$M_t = r \cdot G; \quad r = \frac{4,472 \text{ Nm}}{200 \text{ N}} = 0,0224 \text{ m} \quad (10)$$



$$1) \quad r = 6 \text{ cm}$$

$$l = 10 \text{ cm}$$

$$G = 120 \text{ N}$$

$$a = r \cdot \sqrt{2} = 8,485 \text{ cm} \quad (5)$$

$$\varphi = \arcsin \frac{a}{l+r} = \arcsin \frac{8,485}{16}$$

$$\varphi = 32,03^\circ \quad (5)$$

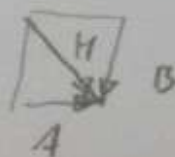
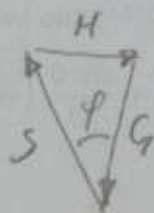
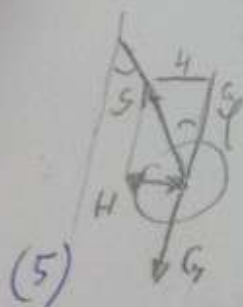
$$\cos \varphi = \frac{G}{S}; \quad S = \frac{G}{\cos \varphi} = \frac{120 \text{ N}}{\cos 32,03^\circ}$$

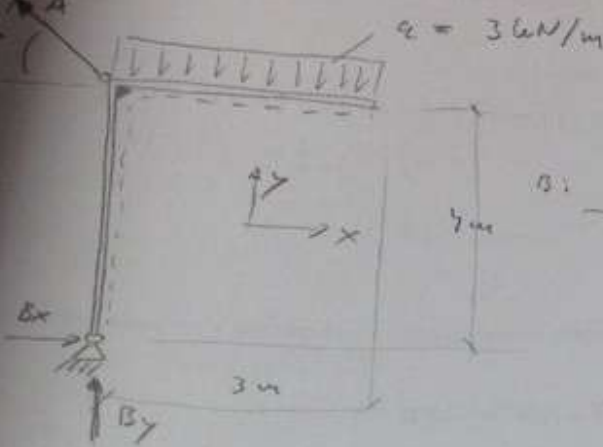
$$S = 141,5 \text{ N} \quad (5)$$

$$H = G \cdot \tan \varphi; \quad H = 120 \cdot \tan 32,03^\circ$$

$$H = 75,06 \text{ N} \quad (5)$$

$$A = B = H \cdot \frac{\sqrt{2}}{2} = 53,08 \text{ N} \quad (5)$$

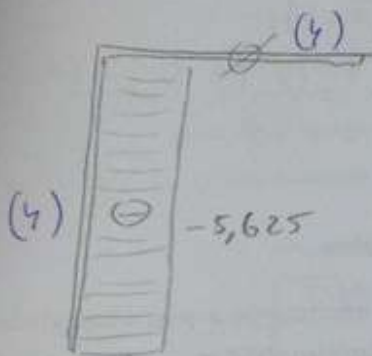




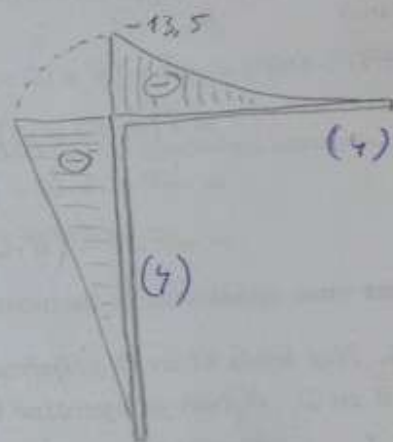
$$\begin{aligned} -A \cdot \cos 45^\circ + B_x &= 0 \\ 4 \cdot \sin 45^\circ + B_y - 2 \cdot 3 &= 0 \\ B: -2 \cdot 3 \cdot 1,5 + A \cdot \cos 45^\circ \cdot 4 &= 0 \end{aligned}$$

$$\begin{aligned} A &= 4,773 \text{ kN} \\ B_x &= 3,375 \text{ kN} \\ B_y &= 5,625 \text{ kN} \end{aligned} \quad (11)$$

$N[\text{kN}]$



$M[\text{kNm}]$



$T[\text{kN}]$

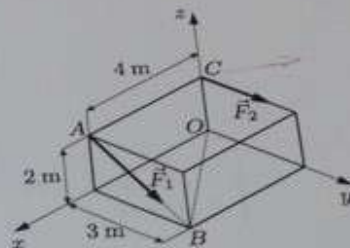


3. IZPIT IZ STATIKE
NTF Geotehnologija in rudarstvo UN
30. januar 2012

Čas pisanja je 90 min. Tega lista ne oddajate. Skice, ki definirajo nastopajoče veličine, sodijo v vaš izdelek. V le-tém naj bo razviden postopek reševanja.

- (1) Za prikazani sistem sil velja: $|\vec{F}_1| = 3 \text{ kN}$, $|\vec{F}_2| = 7 \text{ kN}$.

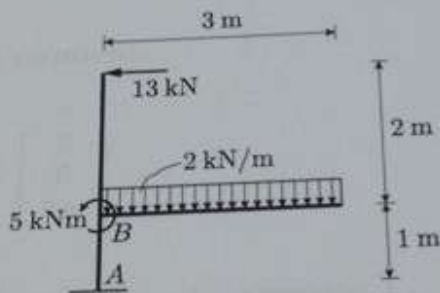
- (a) Izvedite redukcijo prikazanega sistema sil v koordinatno izhodišče O .
(b) Določite, s kolikšnim momentom deluje ta sistem sil okrog osi OB .



- (2) Za konstrukcijo na sliki:

- (a) Izračunajte reakcije v podpori A .
(b) Izrišite diagrame notranje osne sile, prečne sile in notranjega upogibnega momenta.
(c) Določite maksimalni upogibni moment.

Priporočljiva je izvedba kontrole ravnovesja momentov v točki B .



Redukcija

$$\vec{R}_O = \vec{F}_1 + \vec{F}_2 \quad \vec{M}_O = \vec{M}_1 + \vec{M}_2 = \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 \quad (1)$$

1. Rezultanta

$$\vec{F}_1 = 3 \text{ kN} \cdot \vec{e}_1, \quad \vec{e}_1 = \begin{bmatrix} 0 \\ 3 \\ -2 \end{bmatrix} \cdot \frac{1}{\sqrt{2^2 + (-3)^2}} = \begin{bmatrix} 0 \\ 0,832 \\ -0,555 \end{bmatrix}, \quad \vec{F}_1 = \begin{bmatrix} 0 \\ 2,496 \text{ kN} \\ -1,664 \text{ kN} \end{bmatrix} \quad (2)$$

$$\vec{F}_2 = \begin{bmatrix} 0 \\ 7 \text{ kN} \\ 0 \end{bmatrix} \quad (3)$$

$$\vec{R}_O = \begin{bmatrix} 0 \\ 9,496 \text{ kN} \\ -1,664 \text{ kN} \end{bmatrix} \quad (4)$$

1.2 Rezultirajoči moment

$$\vec{r}_1 = \begin{bmatrix} 4 \text{ m} \\ 0 \\ 2 \text{ m} \end{bmatrix}, \quad \vec{r}_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \text{ m} \end{bmatrix} \quad (5)$$

$$\vec{M}_1 = \begin{bmatrix} 4 \text{ m} \\ 0 \\ 2 \text{ m} \end{bmatrix} \times \begin{bmatrix} 0 \\ 2,496 \text{ kN} \\ -1,664 \text{ kN} \end{bmatrix} = \begin{bmatrix} -4,992 \text{ kNm} \\ 6,656 \text{ kNm} \\ 9,984 \text{ kNm} \end{bmatrix} \quad (6)$$

$$\vec{M}_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \text{ m} \end{bmatrix} \times \begin{bmatrix} 0 \\ 7 \text{ kN} \\ 0 \end{bmatrix} = \begin{bmatrix} -14 \text{ kNm} \\ 0 \\ 0 \end{bmatrix} \quad (7)$$

$$\vec{M}_O = \begin{bmatrix} -18,99 \text{ kNm} \\ 6,656 \text{ kNm} \\ 9,984 \text{ kNm} \end{bmatrix}$$

3 Določitev momenta M_{OB}

$$M_{OB} = \vec{M}_O \vec{e}_{OB} = \begin{bmatrix} -18,99 \text{ kNm} \\ 6,656 \text{ kNm} \\ 9,984 \text{ kNm} \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix} \cdot \frac{1}{\sqrt{4^2 + 3^2}} \quad (9)$$

$$= \begin{bmatrix} -18,99 \text{ kNm} \\ 6,656 \text{ kNm} \\ 9,984 \text{ kNm} \end{bmatrix} \cdot \begin{bmatrix} 0,8 \\ 0,6 \\ 0 \end{bmatrix} = \boxed{-11,2 \text{ kNm}} \quad (10)$$

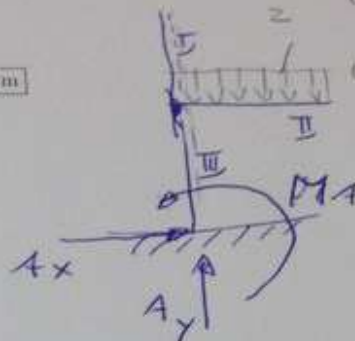
2 Določitev notranjih veličin

2.1 Reakcije

$$x: A_x - 13 \text{ kN} = 0 \Rightarrow A_x = \boxed{13 \text{ kN}} \quad (11)$$

$$y: A_y - 2 \text{ kN/m} \cdot 3 \text{ m} = 0 \Rightarrow A_y = \boxed{6 \text{ kN}} \quad (12)$$

$$M_A: M_A + 5 \text{ kNm} + 13 \text{ kN} \cdot 3 \text{ m} - 2 \text{ kN/m} \cdot 3 \text{ m} \cdot 1,5 \text{ m} = 0 \Rightarrow M_A = \boxed{-35 \text{ kNm}} \quad (13)$$



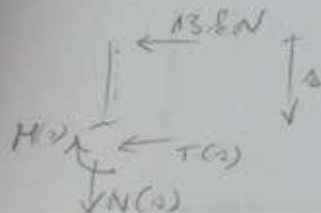
2.2 Preračun posameznih polj

I. $s \in [0, 2 \text{ m}]$

$$N(s) = 0$$

$$T(s) = -13 \text{ kN}$$

$$M(s) = \boxed{13 \text{ kN} \cdot s}$$

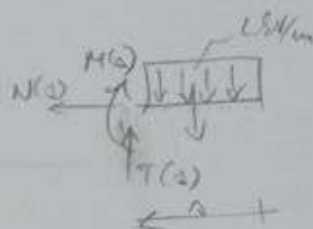


II. $s \in [0, 3 \text{ m}]$

$$N(s) = 0$$

$$T(s) = 2 \text{ kN/m} \cdot s$$

$$M(s) = \boxed{-2 \text{ kN/m} \cdot \frac{s^2}{2}}$$



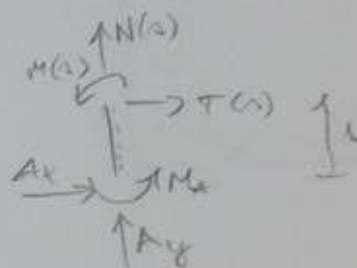
III. $s \in [0, 1 \text{ m}]$

$$N(s) = -A_y = -6 \text{ kN}$$

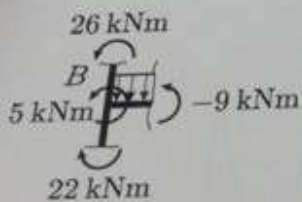
$$T(s) = -A_x = -13 \text{ kN}$$

$$M(s) = -M_A - A_x \cdot s$$

$$M(s) = \boxed{35 \text{ kNm} - 13 \text{ kN} \cdot s}$$



2.3 Kontrola momentov v B



$$26 \text{ kNm} + (-9 \text{ kNm}) - 22 \text{ kNm} + 5 \text{ kNm} = 0 \quad (14)$$

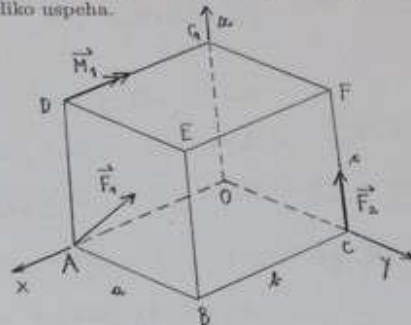
1. IZPIT IZ STATIKE

NTF Geotehnologija VSR in Geotehniška tehnologija VSR
29. januar 2009

UNI (Bologna št.)

Iz rešitve ustrezno utemeljite. Iz vašega izdelka naj bo razviden postopek reševanja. Naloge so redne 33+34+33 točk. Profesor in asistent vam želita veliko uspeha.

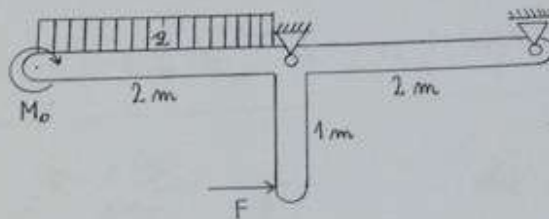
- (1) Na kvader z dimenzijami $a = 2.4 \text{ m}$, $b = 3 \text{ m}$ in $c = 3.2 \text{ m}$ deluje sistem dveh sil in momenta. Sila $\vec{F}_1$ deluje v točki A in kaže proti E, njena velikost pa je 8 kN , sila $\vec{F}_2$ pa ima velikost 4 kN in prijemlje v točki C. Moment $\vec{M}_1$ deluje v točki D, njegova velikost pa je 5 kNm . Reduciraj ta sistem v točko B.



- (2) Za nosilec na sliki

- izračunaj reakcije v podporah,
- nariši diagrame osne in prečne sile ter upogibnega momenta,
- določi maksimalni upogibni moment in mesto, kjer nastopi.

Pri tem je sila $F = 5 \text{ kN}$, točkasti moment $M_0 = 7 \text{ kNm}$ in linearna gostota sile $q = 2 \frac{\text{kN}}{\text{m}}$.



- 1) Za narisano parabolično vrv izračunajte h , b , S_{max}

